BY: MR. NEGUSU GEBRMICHAEL

STROKE PREDICTION DATASET

DATE: DEC 21,2022

INSTRUCTOR: MR. ADRIAN ARULSEELAN
METRO COLLEGE OF TECHNOLOGY, TORONTO, CANADA

OUTLINE

Mr. Negusu GEBRMICHAEL

- Introduction
- Data Info
- Library
- Data Cleaning
- Normality Test

- Univariate
- Bivariate

- Correlation
- T-test
- Chi-square

- Regression
- Multivariate
- Logistic Regression
- Conclusions

INTRODUCTION

https://www.kaggle.com/datasets/fedesoriano/stroke-prediction-dataset

According To The World Health Organization (WHO) Stroke Is The 2nd Leading Cause Of Death Globally, Responsible For Approximately 11% Of Total Deaths.

This Dataset Is Used To Predict Whether A Patient Is Likely To Get Stroke Based On The Input Parameters Like Gender, Age, Various diseases, and smoking status. Each row in the data provides relevant information about the

Column	Description
id	unique identifier
gender	"Male", "Female" or "Other"
age	age of the patient
hypertension	0 = none-hypertension, 1 = hypertension
heart_disease	0 = none- heart diseases, 1 = heart disease
ever_married	"No" or "Yes"
work_type	“Children", "Govt_jov", "Never_worked", "Private" or "Self-employed"
Residence_type	"Rural" or "Urban"
avg_glucose_level	average glucose level in blood
bmi	body mass index
smoking_status	"formerly smoked", "never smoked", "smokes" or "Unknown"*

DATA INFO

```
dataset = pd.read_csv('healthcare-dataset-stroke-data.csv')
```

```
dataset.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 5110 entries, 0 to 5109
Data columns (total 12 columns):
 #   Column             Non-Null Count  Dtype
---  ------             --------------  -----
 0   id                 5110 non-null   int64
 1   gender             5110 non-null   object
 2   age                5110 non-null   float64
 3   hypertension       5110 non-null   int64
 4   heart_disease      5110 non-null   int64
 5   ever_married       5110 non-null   object
 6   work_type          5110 non-null   object
 7   Residence_type     5110 non-null   object
 8   avg_glucose_level  5110 non-null   float64
 9   bmi                4909 non-null   float64
 10  smoking_status     5110 non-null   object
 11  stroke             5110 non-null   int64
dtypes: float64(3), int64(4), object(5)
memory usage: 479.2+ KB
```

```
import numpy as np
dataset.describe()
```

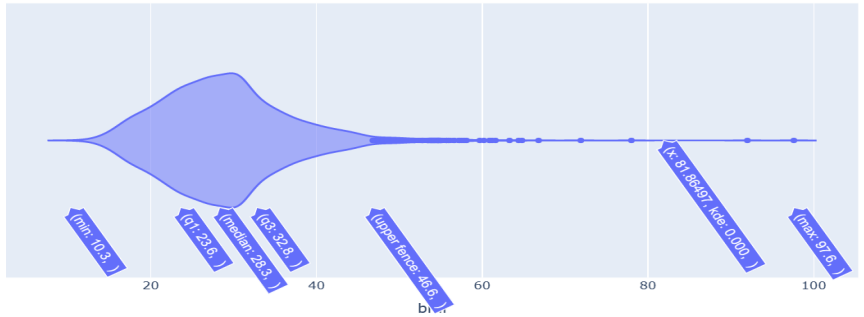
	id	age	hypertension	heart_disease	avg_glucose_level	bmi	stroke
count	5110.000000	5110.000000	5110.000000	5110.000000	5110.000000	4909.000000	5110.000000
mean	36517.829354	43.226614	0.097456	0.054012	106.147677	28.893237	0.048728
std	21161.721625	22.612647	0.296607	0.226063	45.283560	7.854067	0.215320
min	67.000000	0.080000	0.000000	0.000000	55.120000	10.300000	0.000000
25%	17741.250000	25.000000	0.000000	0.000000	77.245000	23.500000	0.000000
50%	36932.000000	45.000000	0.000000	0.000000	91.885000	28.100000	0.000000
75%	54682.000000	61.000000	0.000000	0.000000	114.090000	33.100000	0.000000
max	72940.000000	82.000000	1.000000	1.000000	271.740000	97.600000	1.000000

LIBRARY

```
# library
import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt
%matplotlib inline
import math
import statsmodels.api as sm
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression,LogisticRegression
from sklearn import metrics
from statsmodels.graphics.gofplots import qqplot
from sklearn.preprocessing import LabelEncoder
from sklearn.preprocessing import PowerTransformer
import plotly.express as px
import scipy.stats as stats
import os
from sklearn.metrics import classification_report, confusion_matrix
from sklearn.metrics import classification_report,confusion_matrix,f1_score
from sklearn.metrics import classification_report
from scipy.stats import chi2_contingency
import warnings
warnings.filterwarnings('ignore')
import pingouin as pg
from sklearn.pipeline import Pipeline
from sklearn.svm import SVC
```

DATA CLEANING- OUTLIERS

```
import numpy as np
import plotly.express as px
px.line(data_frame=df,x='age' ,y='bmi' ,title="Lets view our data for outliers")
px.violin(data_frame=df, x='bmi')
```



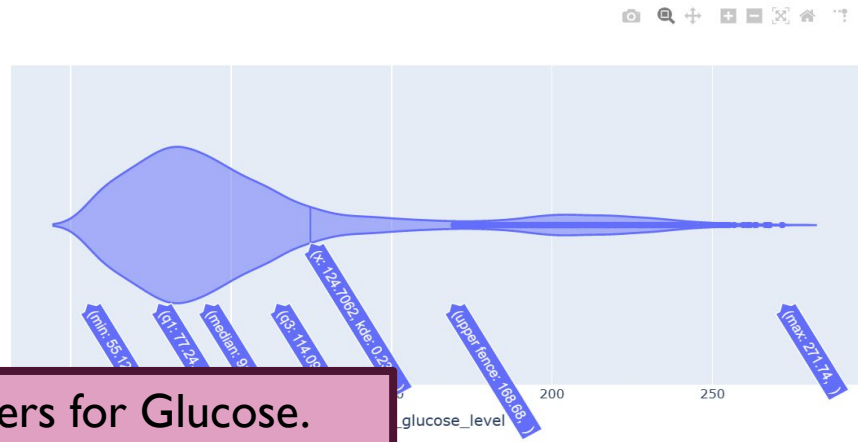

```
def traditional_outlier(df,x):
    q1 = df[x].quantile(.25)
    q3 = df[x].quantile(.75)
    iqr=q3-q1
    df['Traditional'] =np.where(df[[x]]<(q1-1.5*iqr),-1,
                                np.where(df[[x]]>(q3+1.5*iqr),-1,1))
    return df

traditional_outlier(df,'bmi')
```

	gender	age	hypertension	heart_disease	ever_married	work_type	Residence_type	avg_glucose_level	bmi	smoking_status	stroke	age_range	Traditional
0	Male	67.0	0	1	Yes	Private	Urban	228.69	36.600000	formerly smoked	1	[20, 85)	1
1	Female	61.0	0	0	Yes	Self-employed	Rural	202.21	30.541211	never smoked	1	[20, 85)	1
2	Male	80.0	0	1	Yes	Private	Rural	105.92	32.500000	never smoked	1	[20, 85)	1
3	Female	49.0	0	0	Yes	Private	Urban	171.23	34.400000	smokes	1	[20, 85)	1
4	Female	79.0	1	0	Yes	Self-employed	Rural	174.12	24.000000	never smoked	1	[20, 85)	1
...	...	...	...	...	...	...	...	...	...	...	...	...	...
5105	Female	80.0	1	0	Yes	Private	Urban	83.75	30.541211	never smoked	0	[20, 85)	1
5106	Female	81.0	0	0	Yes	Self-employed	Urban	125.20	40.000000	never smoked	0	[20, 85)	1
5107	Female	35.0	0	0	Yes	Self-employed	Rural	82.99	30.600000	never smoked	0	[20, 85)	1
5108	Male	51.0	0	0	Yes	Private	Rural	166.29	25.600000	formerly smoked	0	[20, 85)	1
5109	Female	44.0	0	0	Yes	Govt_job	Urban	85.28	26.200000	Unknown	0	[20, 85)	1

5110 rows × 13 columns

121 Outliers for BMI and 627 Outliers for Glucose.

```
import numpy as np
import plotly.express as px
px.line(data_frame=df,x='age' ,y='avg_glucose_level' ,title="Lets view our data for outliers")
px.violin(data_frame=df, x='avg_glucose_level')
```




```
traditional_outlier(df,'avg_glucose_level')
```

	gender	age	hypertension	heart_disease	ever_married	work_type	Residence_type	avg_glucose_level	bmi	smoking_status	stroke	age_range	Traditional
0	Male	67.0	0	1	Yes	Private	Urban	228.69	36.600000	formerly smoked	1	[20, 85)	-1
1	Female	61.0	0	0	Yes	Self-employed	Rural	202.21	30.541211	never smoked	1	[20, 85)	-1
2	Male	80.0	0	1	Yes	Private	Rural	105.92	32.500000	never smoked	1	[20, 85)	1
3	Female	49.0	0	0	Yes	Private	Urban	171.23	34.400000	smokes	1	[20, 85)	-1
4	Female	79.0	1	0	Yes	Self-employed	Rural	174.12	24.000000	never smoked	1	[20, 85)	-1
...	...	...	...	...	...	...	...	...	...	...	...	...	...
5105	Female	80.0	1	0	Yes	Private	Urban	83.75	30.541211	never smoked	0	[20, 85)	1
5106	Female	81.0	0	0	Yes	Self-employed	Urban	125.20	40.000000	never smoked	0	[20, 85)	1
5107	Female	35.0	0	0	Yes	Self-employed	Rural	82.99	30.600000	never smoked	0	[20, 85)	1
5108	Male	51.0	0	0	Yes	Private	Rural	166.29	25.600000	formerly smoked	0	[20, 85)	1
5109	Female	44.0	0	0	Yes	Govt_job	Urban	85.28	26.200000	Unknown	0	[20, 85)	1

5110 rows × 13 columns

DATA CLEANING- UNIQUE VALUE & MISSING VALUE

(1) Drop ID and Replace Other Gender with Mode

```
df=dataset.drop(columns=['id'], inplace=False)
```

```
dataset['gender'].value_counts()
```

```
Female    2994
Male      2115
Other        1
Name: gender, dtype: int64
```

```
print(dataset['gender'].mode())
dataset['gender'].replace('Other','Female', inplace=True)
```

```
0    Female
dtype: object
```

```
dataset['gender'].value_counts()
```

```
Female    2995
Male      2115
Name: gender, dtype: int64
```

(2) Filling Missing Values with 3 Mean BMI Age-wise

```
df['bmi'].isna().mean() * 100
```

```
0.0
```

```
df["age_range"] = pd.cut(df.age, bins=[0,2,20,85], right=False)


mean_df=df.groupby(["age_range"]).bmi.mean()
print(mean_df)
means=mean_df.agg(list) # converts all 3 means to python list
for index,row in df.iterrows():

    age=row["age"]
    if age>=0 and age<2:

       row.replace(np.nan,means[0],inplace=True)

    elif age>=2 and age<20:
        row.replace(np.nan,means[1],inplace=True)

    else:
        row.replace(np.nan,means[2],inplace=True)
    df.loc[ index] =row
```

```
age_range
[0, 2)      18.478070
[2, 20)     22.451205
[20, 85)    30.541211
Name: bmi, dtype: float64
```

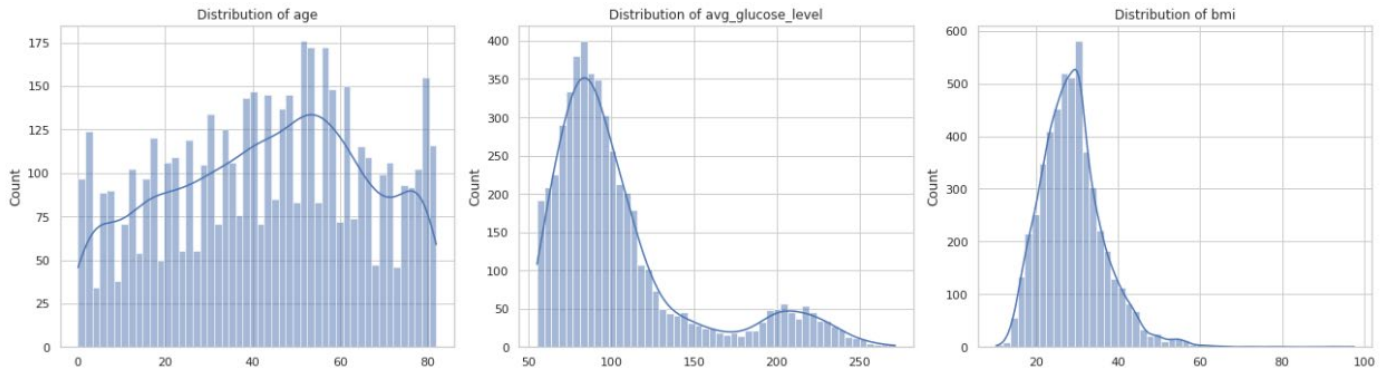
NORMALITY TEST

1st Step : Check the Normality

```
# defining numerical variables in a dataframe num_at
num_attribute = ['age', 'avg_glucose_level', 'bmi']
df_nums = df[num_attribute]
df_nums.head()
```

	age	avg_glucose_level	bmi
3295	0.08	70.33	16.9
1614	0.08	139.67	14.1
3618	0.16	114.71	17.4
4021	0.16	109.52	13.9
3968	0.16	69.79	13.0

```
#Layer 1 Normality Test
# Statistical Analysis on Contineous Numerical Features Asma

fig = plt.figure(1, (18, 5))

for i, attribute in enumerate(num_attribute):
    ax = plt.subplot(1, 3, i+1)
    # sns.displot(data=df_nums, x=attribute, kde=True)
    sns.histplot(data=df_nums, x=attribute, kde=True, bins=50)
    ax.set_xlabel(None)
    ax.set_title(f'Distribution of {attribute}')
    plt.tight_layout()
plt.show()
# IF you see the graph below , only age is normaly distributed. It can be seen that indeed the distribution is very skewed to the le
```



NORMALITY TEST

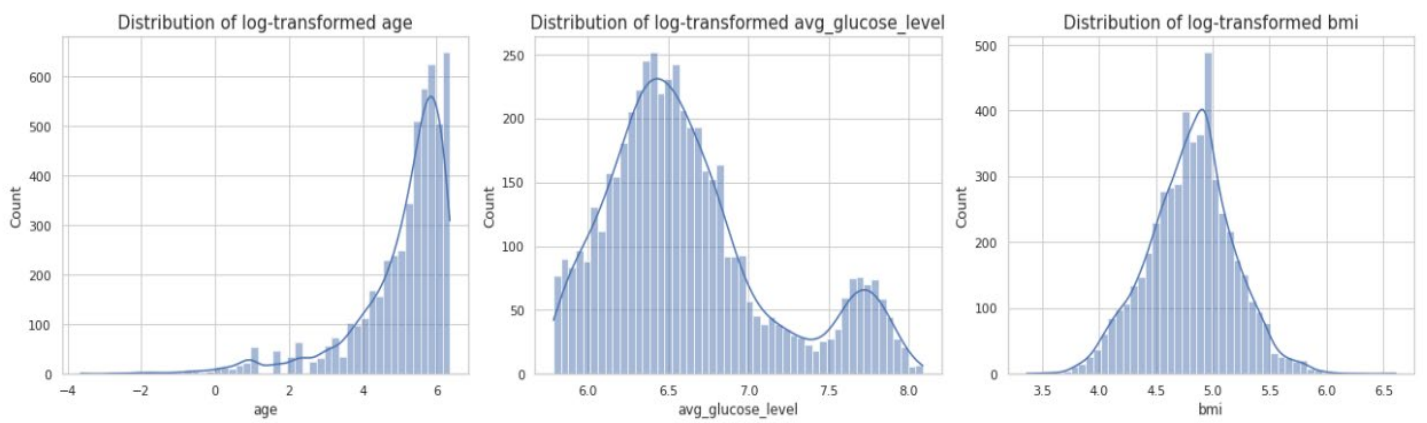
2nd Step : Check the Log-normal

```
#A log-normal distribution is a distribution of a random variable whose
# Checking whether these features are log-normal or not.

df_log = np.log2(df_nums)
df_log.head()
```

	age	avg_glucose_level	bmi
3295	-3.643856	6.136068	4.078951
1614	-3.643856	7.125878	3.817623
3618	-2.643856	6.841847	4.121015
4021	-2.643856	6.775051	3.797013
3968	-2.643856	6.124948	3.700440

1.3s

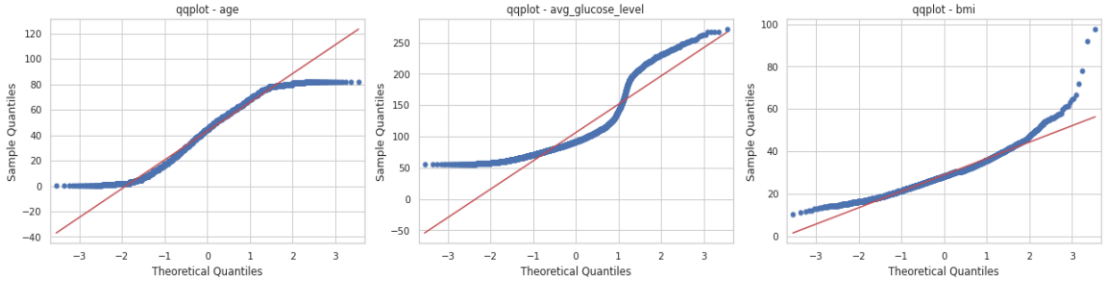
```
fig = plt.figure(1, (18, 5))

for i, attribute in enumerate(num_attribute):
    ax = plt.subplot(1, 3, i+1)
    ax.set_title(f'Distribution of log-transformed {attribute}', size=15)
    sns.histplot(data=df_log, x=attribute, kde=True, bins=50)
    plt.tight_layout()
plt.show()

# age is left skewed, while avg_glucose_level is also not log normal
# bmi has become almost entirely normal distribution after the log transformation.
# Higher peak iin the middle is because mean imputation
```



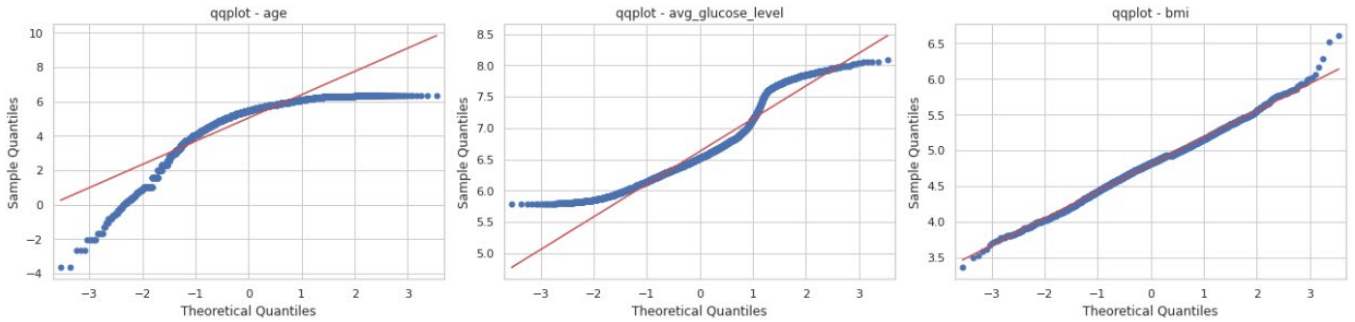
NORMALITY TEST

3rd Step : Log Transformation

```
from statsmodels.graphics.gofplots import qqplot
# quantile-quantile plots on original data
fig = plt.figure(1, (18, 8))

for i, attribute in enumerate(df_nums):
    ax = plt.subplot(2, 3, i+1)
    qqplot(df_nums[attribute], line='s', ax=ax, markersize=5)
    ax.set_title(f'qqplot - {attribute}')
    plt.tight_layout()
plt.show()

#visualization before transformation
#age is similar to before transformation, while avg_glucose_level has become more normal-like but still far from ideal.
#bmi has become almost entirely normal distribution after the power transformation.
```




```
fig = plt.figure(1, (18,8))

for i, num_feature in enumerate(df_nums):
    ax = plt.subplot(2,3,i+1)
    sm.qqplot(df_log[num_feature], line='s', ax=ax, markersize=5)
    ax.set_title(f'qqplot - {num_feature}')
    plt.tight_layout()
plt.show()

# visualization after applying log transformation
```



NORMALITY TEST

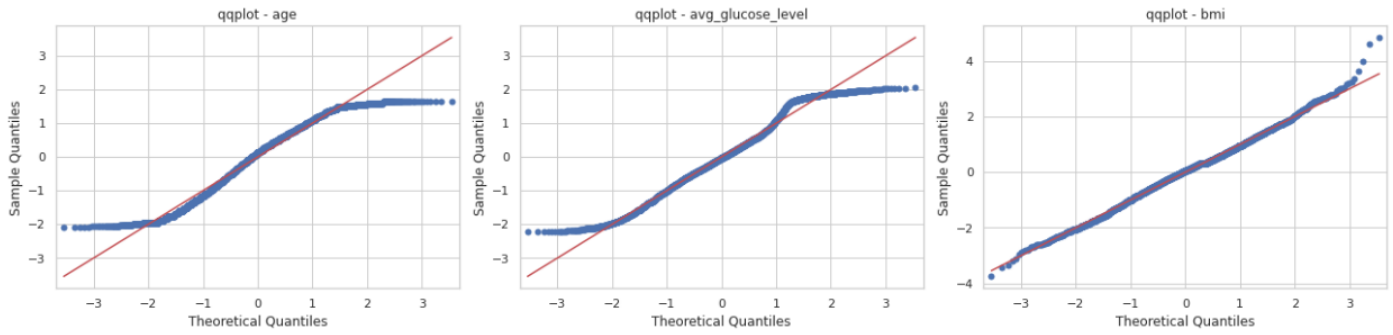
4th Step : Power Transformation

```
# Power transformer by defaul takes yeo-johnson method to make distribut
from sklearn.preprocessing import PowerTransformer

pt = PowerTransformer()
transformed_df = pt.fit_transform(df_nums)
transformed_df = pd.DataFrame(transformed_df, columns=df_nums.columns)
transformed_df.head()
```

Table Raw Visualize Statistics

	age	avg_glucose_...	bmi
0	-2.07926422980449	-1.033592581228881	-1.91696012881604…
1	-2.07926422980449	1.0636591548902505	-2.594946743020776
2	-2.073509203929598	0.6079989337891626	-1.80721138514531…
3	-2.073509203929598	0.486324685327995	-2.648115135903927
4	-2.073509203929598	-1.066805499151809	-2.89660504355823…

```
fig = plt.figure(1, (18,8))

for i, num_feature in enumerate(df_nums):
    ax = plt.subplot(2,3,i+1)
    sm.qqplot(transformed_df[num_feature], line='s', ax=ax, markersize=5)
    ax.set_title(f'qqplot - {num_feature}')
    plt.tight_layout()
plt.show()

# Visualization after applying power transformation
```



UNIVARIATE ANALYSIS

Change Binominal Categorical to Numerical

```
df['gender'].replace(['Male', 'Female'],
                     [0, 1], inplace=True)
df['ever_married'].replace(['No', 'Yes'],
                     [0, 1], inplace=True)
df['work_type'].replace(['Never_worked', 'Govt_job', 'children', 'Self-employed', 'Private'],
                     [0, 1, 2, 3, 4], inplace=True)
df['Residence_type'].replace(['Rural','Urban'],
                     [0, 1], inplace=True)
df['smoking_status'].replace(['smokes', 'formerly smoked', 'Unknown', 'never smoked'],
                     [0, 1, 2, 3], inplace=True)

print(df)
```

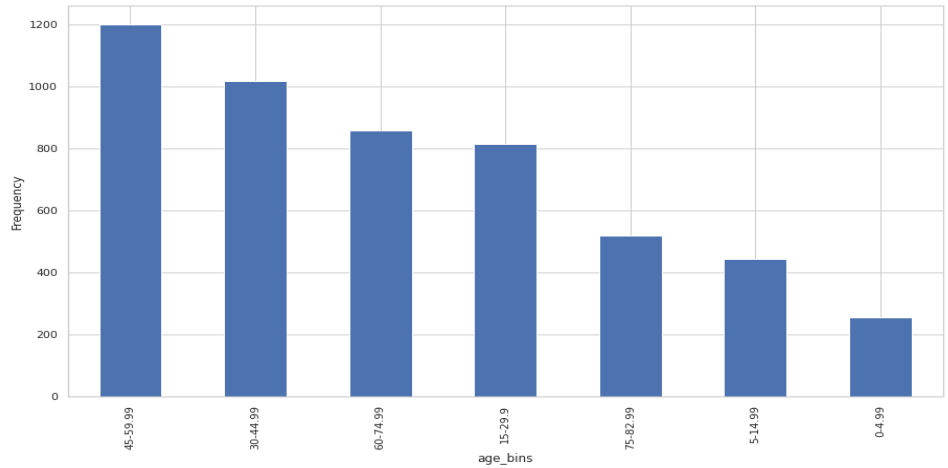
+ Show all

```
4094              1            111.81  19.8              1       0
2341              1             80.00  33.6              3       0
4716              0             96.98  21.5              3       0
187               1            215.94  27.9              1       1

     age_range  Traditional
3295    [0, 2)            1
1614    [0, 2)            1
3618    [0, 2)            1
4021    [0, 2)            1
3968    [0, 2)            1
...        ...          ...
4590  [20, 85)            1
4094  [20, 85)            1
2341  [20, 85)            1
4716  [20, 85)            1
187   [20, 85)           -1

[5110 rows x 13 columns]
```

Age_bins

```
#  Level1 FREQUENCY and PERCENTAGE of the unique age_bins values  Asma, Y
import matplotlib.pyplot as plt
freq = df['age_bins'].value_counts(dropna=False)
percent=df['age_bins'].value_counts(normalize=True)*100
new_df=pd.DataFrame({"frequency":freq,"percent":percent})
print(new_df)


fig, ax = plt.subplots()
freq.plot(ax=ax,kind='bar')
plt.xlabel("age_bins")
plt.ylabel("Frequency")
```

```
          frequency    percent
45-59.99       1201  23.502935
30-44.99       1018  19.921722
60-74.99        858  16.790607
15-29.9         816  15.968689
75-82.99        518  10.136986
5-14.99         444   8.688845
0-4.99          255   4.990215

Text(0, 0.5, 'Frequency')
```

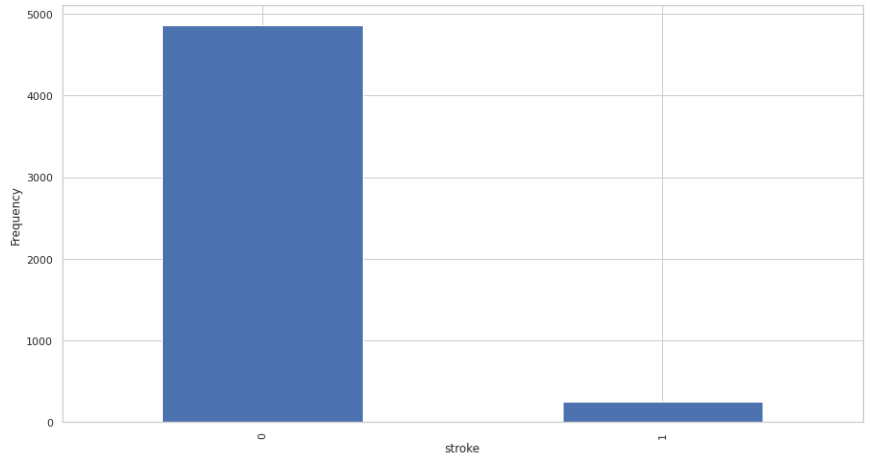


UNIVARIATE ANALYSIS

Stroke

```
#  Level1 FREQUENCY and PERCENTAGE of the unique STROKE values  Asma, Y
import matplotlib.pyplot as plt
freq = df['stroke'].value_counts(dropna=False)
percent=df['stroke'].value_counts(normalize=True)*100
new_df=pd.DataFrame({"frequency":freq,"percent":percent})
print(new_df)


fig, ax = plt.subplots()
freq.plot(ax=ax,kind='bar')
plt.xlabel("stroke")
plt.ylabel("Frequency")
```

```
   frequency    percent
0       4861  95.127202
1        249   4.872798

Text(0, 0.5, 'Frequency')
```

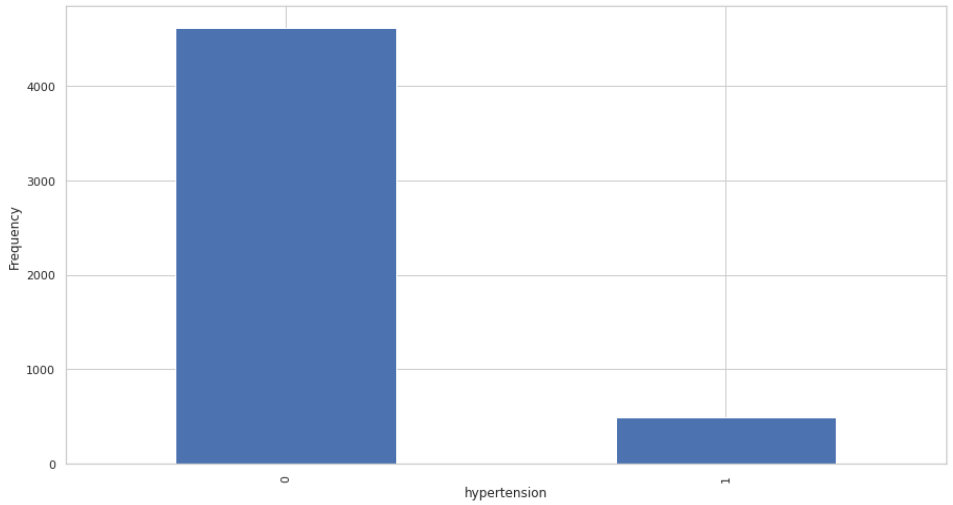


UNIVARIATE ANALYSIS

Hypertension

```
#  Level1 FREQUENCY and PERCENTAGE of the unique hypertension values  Asma, Y
import matplotlib.pyplot as plt
freq = df['hypertension'].value_counts(dropna=False)
percent=df['hypertension'].value_counts(normalize=True)*100
new_df=pd.DataFrame({"frequency":freq,"percent":percent})
print(new_df)


fig, ax = plt.subplots()
freq.plot(ax=ax,kind='bar')
plt.xlabel("hypertension")
plt.ylabel("Frequency")
```

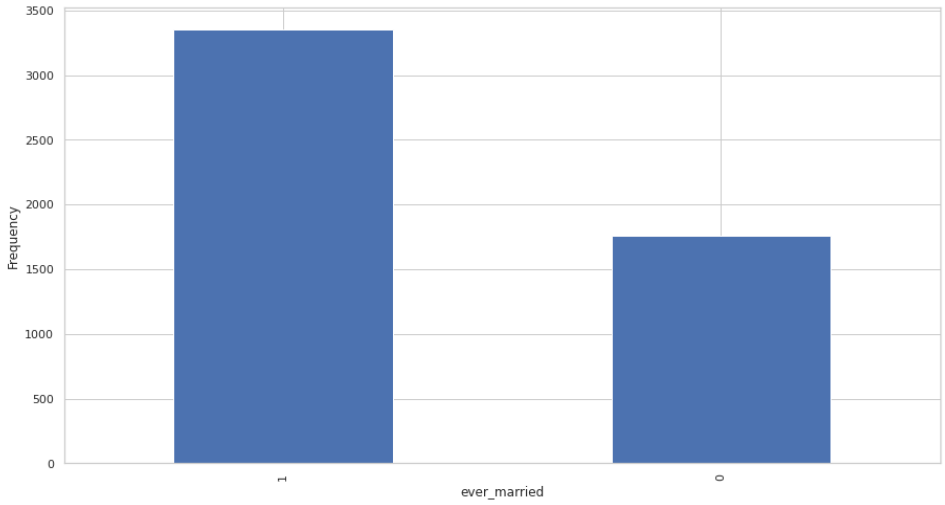
```
   frequency    percent
0       4612  90.254403
1        498   9.745597

Text(0, 0.5, 'Frequency')
```



Ever_married

```
#  Level1 FREQUENCY and PERCENTAGE of the unique hypertension values  Asma, Y
import matplotlib.pyplot as plt
freq = df['ever_married'].value_counts(dropna=False)
percent=df['ever_married'].value_counts(normalize=True)*100
new_df=pd.DataFrame({"frequency":freq,"percent":percent})
print(new_df)


fig, ax = plt.subplots()
freq.plot(ax=ax,kind='bar')
plt.xlabel("ever_married")
plt.ylabel("Frequency")
```

```
   frequency    percent
1       3353  65.616438
0       1757  34.383562

Text(0, 0.5, 'Frequency')
```



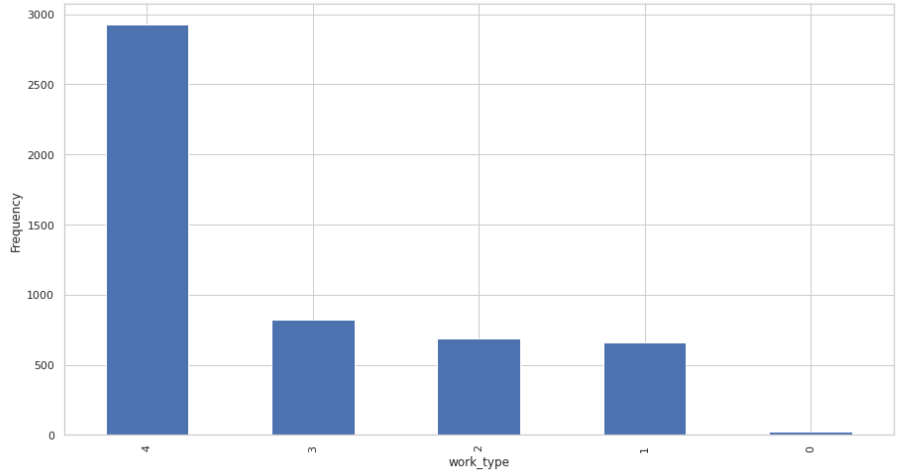
UNIVARIATE ANALYSIS

Work_type

```
#  Level1 FREQUENCY and PERCENTAGE of the unique work_type values  Asma, Y
import matplotlib.pyplot as plt
freq = df['work_type'].value_counts(dropna=False)
percent=df['work_type'].value_counts(normalize=True)*100
new_df=pd.DataFrame({"frequency":freq,"percent":percent})
print(new_df)


fig, ax = plt.subplots()
freq.plot(ax=ax,kind='bar')
plt.xlabel("work_type")
plt.ylabel("Frequency")
```

```
   frequency    percent
4       2925  57.240705
3        819  16.027397
2        687  13.444227
1        657  12.857143
0         22   0.430528
```

```
Text(0, 0.5, 'Frequency')
```

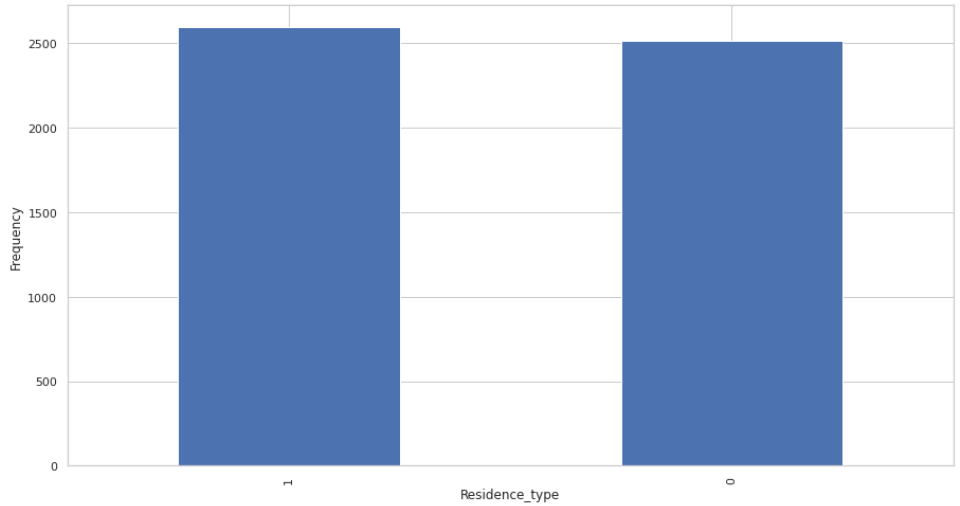


Residence_type

```
#  Level1 FREQUENCY and PERCENTAGE of the unique Residence_type values  Asma, Y
import matplotlib.pyplot as plt
freq = df['Residence_type'].value_counts(dropna=False)
percent=df['Residence_type'].value_counts(normalize=True)*100
new_df=pd.DataFrame({"frequency":freq,"percent":percent})
print(new_df)


fig, ax = plt.subplots()
freq.plot(ax=ax,kind='bar')
plt.xlabel("Residence_type")
plt.ylabel("Frequency")
```

```
   frequency    percent
1       2596  50.802348
0       2514  49.197652
```

```
Text(0, 0.5, 'Frequency')
```

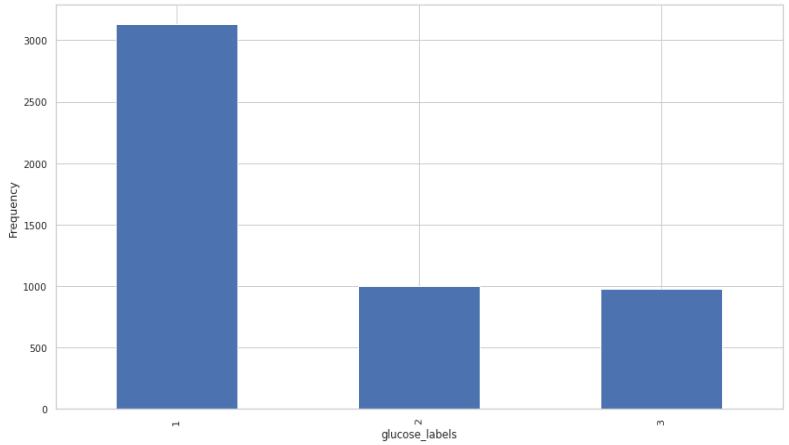


UNIVARIATE ANALYSIS

Glucose_lables

```
#  Level1 FREQUENCY and PERCENTAGE of the unique glucose_labels values  Asma, Y
import matplotlib.pyplot as plt
freq = df['glucose_labels'].value_counts(dropna=False)
percent=df['glucose_labels'].value_counts(normalize=True)*100
new_df=pd.DataFrame({"frequency":freq,"percent":percent})
print(new_df)


fig, ax = plt.subplots()
freq.plot(ax=ax,kind='bar')
plt.xlabel("glucose_labels")
plt.ylabel("Frequency")
```

```
   frequency    percent
1       3131  61.272016
2        998  19.530333
3        981  19.197652

Text(0, 0.5, 'Frequency')
```



Glucose_lables . Min & Max

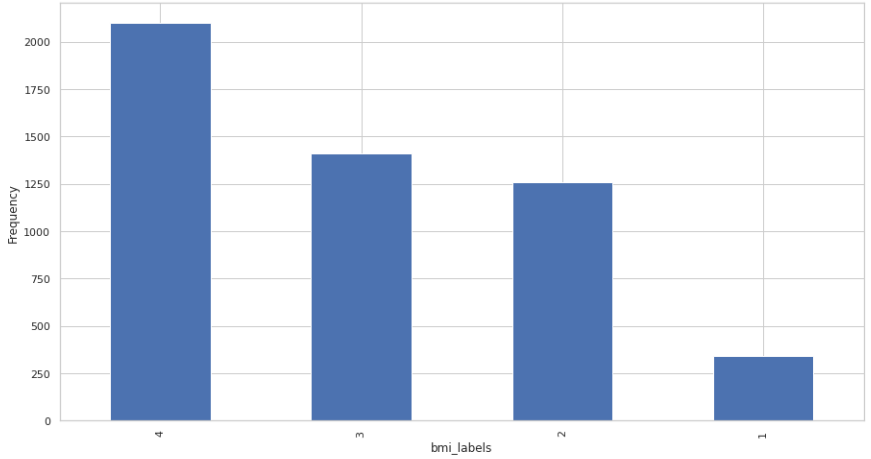
```
#LAYER 1 (Yuliya)
# Finding the max ans min for bining
min_value = df['avg_glucose_level'].min()
max_value = df['avg_glucose_level'].max()
print(min_value)
print(max_value)
```

```
55.12
271.74
```

BMI_lables

```
#  Level1 FREQUENCY and PERCENTAGE of the unique bmi_labels values  Asma, Y
import matplotlib.pyplot as plt
freq = df['bmi_labels'].value_counts(dropna=False)
percent=df['bmi_labels'].value_counts(normalize=True)*100
new_df=pd.DataFrame({"frequency":freq,"percent":percent})
print(new_df)


fig, ax = plt.subplots()
freq.plot(ax=ax,kind='bar')
plt.xlabel("bmi_labels")
plt.ylabel("Frequency")
```

```
   frequency    percent
4       2099  41.076321
3       1409  27.573386
2       1259  24.637965
1        343   6.712329

Text(0, 0.5, 'Frequency')
```




```
#LAYER 1 (Yuliya)
'''BMI  WEIGHT STATUS Binning
Below 18.5  Underweight  1
18.5 - 24.9 Normal       2
25.0 - 29.9 Overweight   3
30.0 and above  Obesity  4
The left bin edge will be exclusive and the right bin edge will be inclus
#LAYER 1 (Yuliya)
# Finding the max ans min for bining
min_value = df['bmi'].min()
max_value = df['bmi'].max()
print(min_value)
print(max_value)
#BMI BINNING 1:10.30-18.49, 2:18.5-24.99, 3:25-29.99 4:30-max
bmi_labels = [1, 2, 3, 4]
cut_bins = [10.20, 18.49, 24.99, 29.99, df["bmi"].max()]
df['bmi_labels'] = pd.cut(df['bmi'], bins=cut_bins, labels=bmi_labels)
```

```
10.3
97.6
```

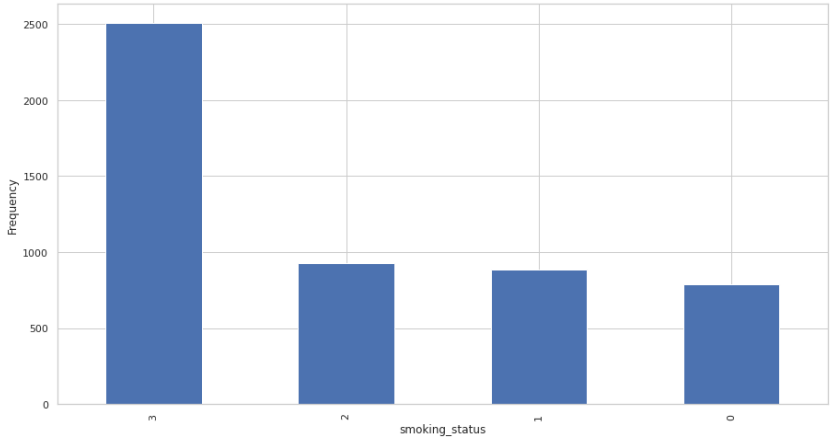
UNIVARIATE ANALYSIS

Smoking_status

```
#  Level1 FREQUENCY and PERCENTAGE of the unique smoking_status values  Asma, Y
import matplotlib.pyplot as plt
freq = df['smoking_status'].value_counts(dropna=False)
percent=df['smoking_status'].value_counts(normalize=True)*100
new_df=pd.DataFrame({"frequency":freq,"percent":percent})
print(new_df)


fig, ax = plt.subplots()
freq.plot(ax=ax,kind='bar')
plt.xlabel("smoking_status")
plt.ylabel("Frequency")
```

```
   frequency    percent
3       2506  49.041096
2        930  18.199609
1        885  17.318982
0        789  15.440313
```

```
Text(0, 0.5, 'Frequency')
```

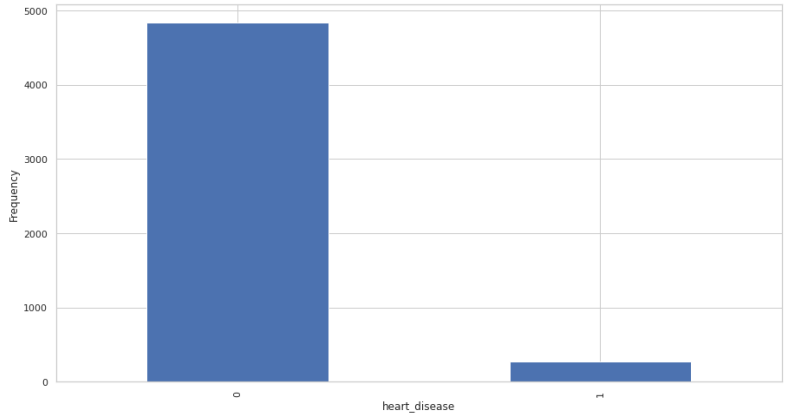


Heart_disease

```
#  Level1 FREQUENCY and PERCENTAGE of the unique HEART_DISEASE values  Asma, Y
import matplotlib.pyplot as plt
freq = df['heart_disease'].value_counts(dropna=False)
percent=df['heart_disease'].value_counts(normalize=True)*100
new_df=pd.DataFrame({"frequency":freq,"percent":percent})
print(new_df)


fig, ax = plt.subplots()
freq.plot(ax=ax,kind='bar')
plt.xlabel("heart_disease")
plt.ylabel("Frequency")
```

```
   frequency    percent
0       4834  94.598826
1        276   5.401174
```

```
Text(0, 0.5, 'Frequency')
```



BIVARIATE ANALYSIS

Stroke vs Gender

```
##Layer 2 Cross Tab Stroke vs Gender (C)
pd.crosstab(df.stroke,df.gender,margins=True)
```

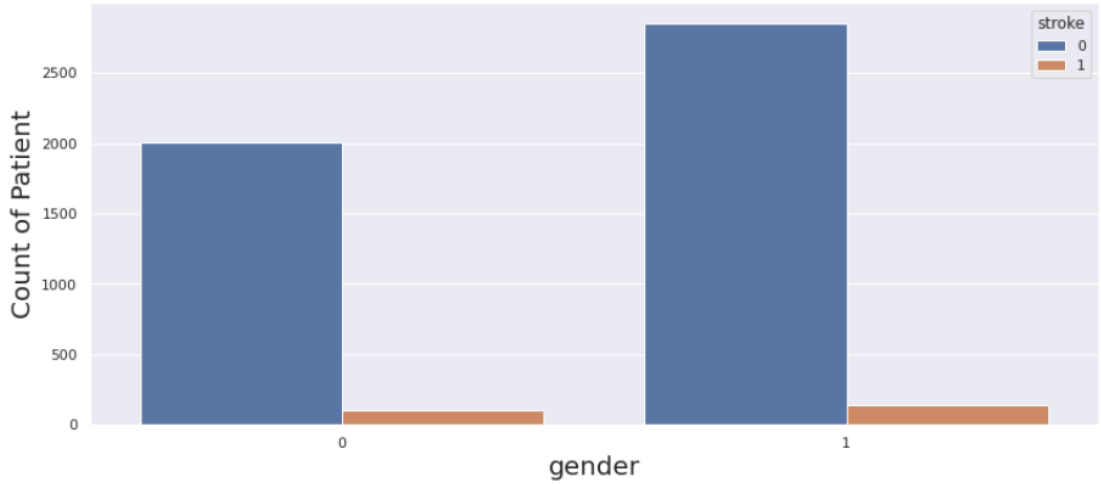
gender	0	1	All
stroke			
0	2007	2854	4861
1	108	141	249
All	2115	2995	5110

```
#Layer 2 Pecentage Cross Tab Stroke vs gender  (C)
pd.crosstab(df.stroke,df.gender, normalize='index')
```

gender	0	1
stroke		
0	0.412878	0.587122
1	0.433735	0.566265

```
#Layer 2 Plot gender (C)
print(df['gender'].value_counts())
sns.set(rc={'figure.figsize':(14,6)})
seaborn_plot = sns.countplot(df['gender'], hue = df['stroke'])
seaborn_plot.set_xlabel("gender",fontsize=20)
seaborn_plot.set_ylabel("Count of Patient",fontsize=20)
# the # of female gender is more than man for the datset so this variable itself is no
```

There may be a certain correlation between the probability of stroke and gender.



BIVARIATE ANALYSIS

Stroke vs Hypertension

```
#Layer 2 Cross Tab Stroke vs hypertension (C)
pd.crosstab(df.stroke,df.hypertension,margins=True)
```

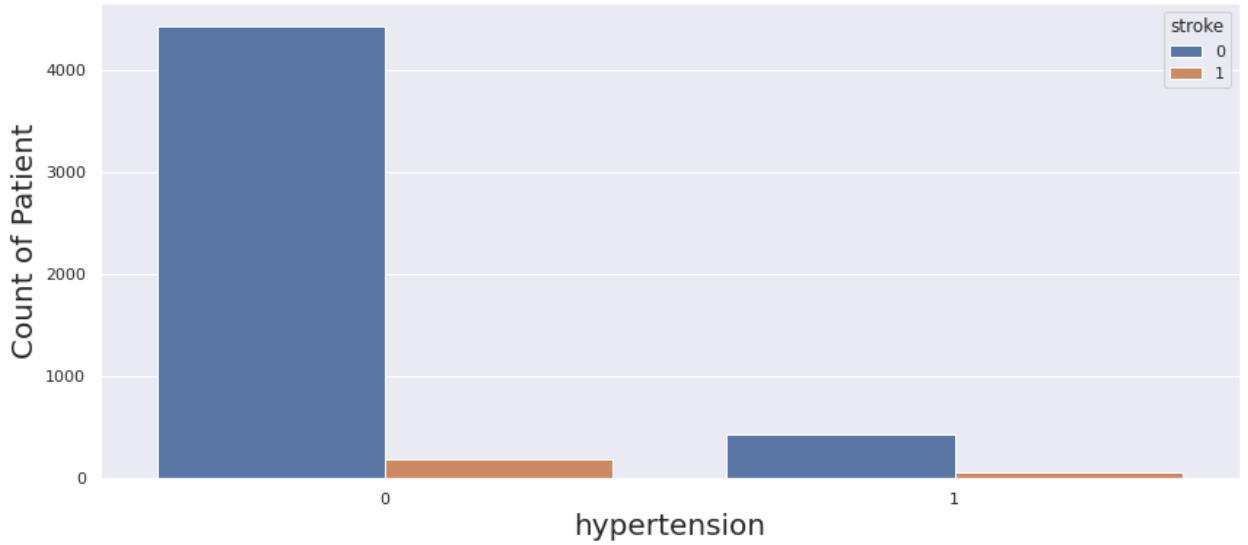
hypertension	0	1	All
stroke			
0	4429	432	4861
1	183	66	249
All	4612	498	5110

```
#Layer 2 Pecentage Cross Tab Stroke vs hypertension  (C)
pd.crosstab(df.stroke,dataset.hypertension, normalize='index')
```

hypertension	0	1
stroke		
0	0.911129	0.088871
1	0.734940	0.265060

```
#Layer 2 Plot hypertension (C)
print(df['hypertension'].value_counts())
sns.set(rc={'figure.figsize':(14,6)})
seaborn_plot = sns.countplot(df['hypertension'], hue = df['stroke'])
seaborn_plot.set_xlabel("hypertension",fontsize=20)
seaborn_plot.set_ylabel("Count of Patient",fontsize=20)
```

```
0    4612
1     498
Name: hypertension, dtype: int64

Text(0, 0.5, 'Count of Patient')
```



BIVARIATE ANALYSIS

Stroke vs Heart_disease

```
#Layer 2 Cross Tab Stroke vs heart_disease  (C)
pd.crosstab(df.stroke,df.heart_disease,margins=True)
```

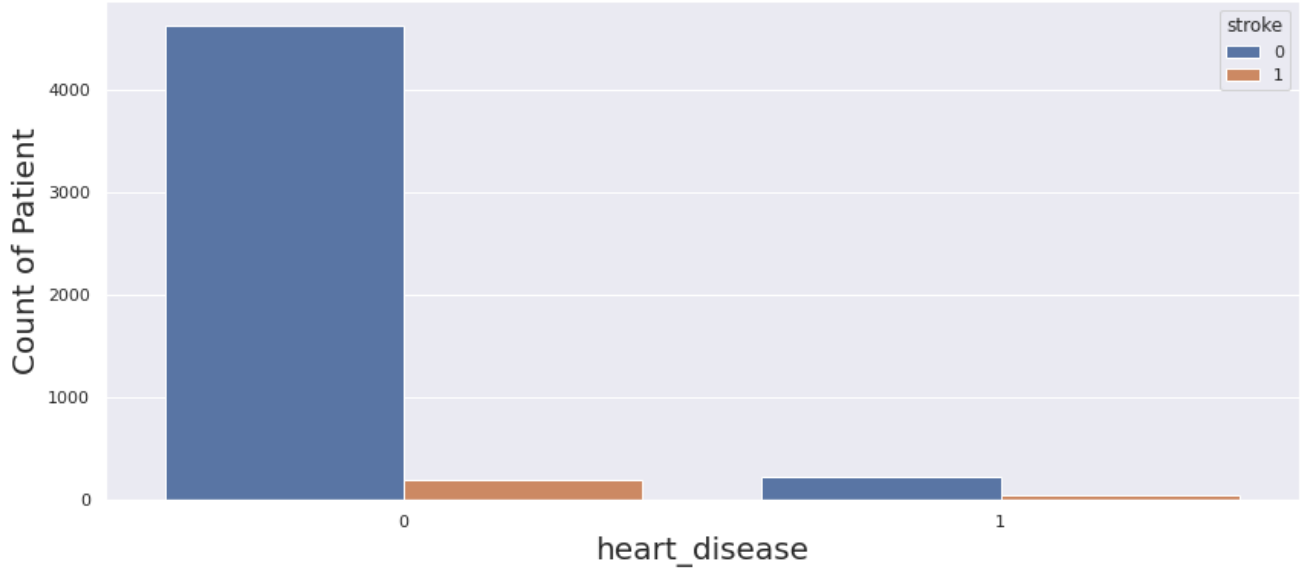
heart_disease	0	1	All
stroke			
0	4632	229	4861
1	202	47	249
All	4834	276	5110

```
#Layer 2 Percentage Cross Tab Stroke vs heart_disease (C)
pd.crosstab(df.stroke,df.heart_disease, normalize='index')
```

heart_disease	0	1
stroke		
0	0.952890	0.047110
1	0.811245	0.188755

```
#Layer 2 Plot heart_disease (C)
print(df['heart_disease'].value_counts())
sns.set(rc={'figure.figsize':(14,6)})
seaborn_plot = sns.countplot(df['heart_disease'], hue = df['stroke'])
seaborn_plot.set_xlabel("heart_disease",fontsize=20)
seaborn_plot.set_ylabel("Count of Patient",fontsize=20)
```

```
0    4834
1     276
Name: heart_disease, dtype: int64

Text(0, 0.5, 'Count of Patient')
```



BIVARIATE ANALYSIS

Stroke vs Ever_married

```
#Layer 2 Cross Tab Stroke vs ever_married  (C)
pd.crosstab(df.stroke,dataset.ever_married,margins=True)
```

ever_married	No	Yes	All
stroke			
0	1728	3133	4861
1	29	220	249
All	1757	3353	5110

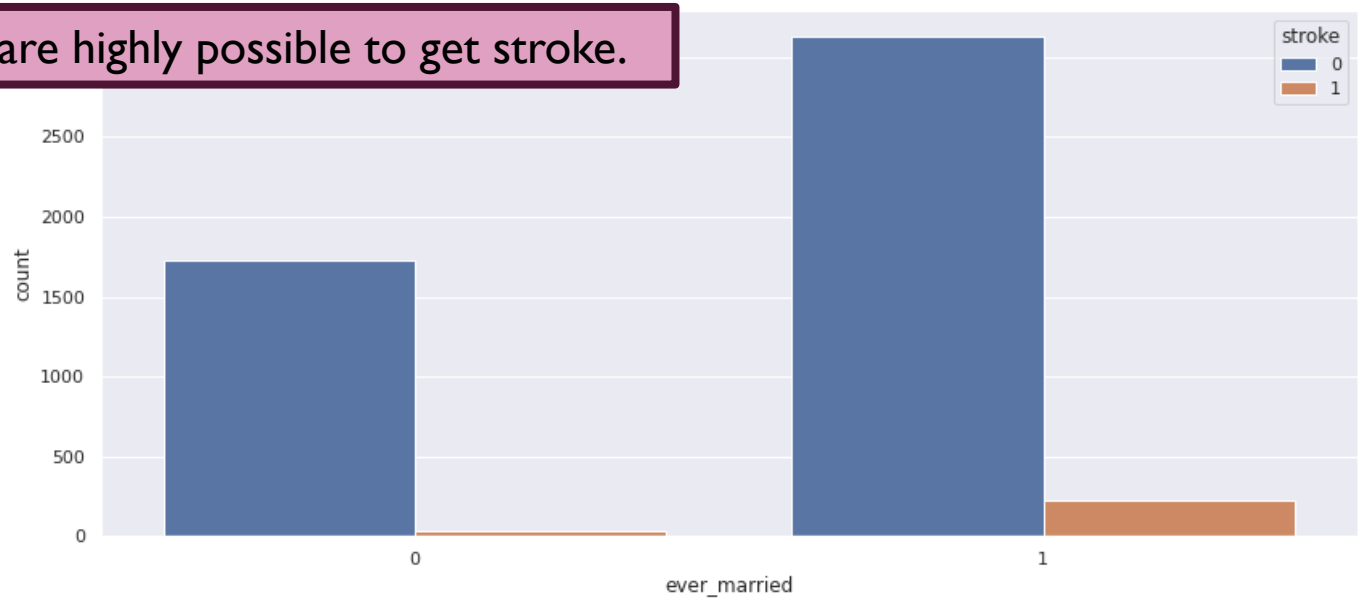
```
#Layer 2 Percentage Cross Tab Stroke vs ever_married  (C)
pd.crosstab(df.stroke,df.ever_married, normalize='index')
```

ever_married	0	1
stroke		
0	0.355482	0.644518
1	0.116466	0.883534

```
#Layer 2 Plot  ever_married  (C)
print(df['ever_married'].value_counts())
sns.countplot(df['ever_married'],hue = df['stroke'])
```

```
1    3353
0    1757
Name: ever_married, dtype: int64

<AxesSubplot:xlabel='ever_married', ylabel='count'>
```

Married people are highly possible to get stroke.



BIVARIATE ANALYSIS

Stroke vs Work_type

```
#Layer 2 Cross Tab Stroke vs work_type (C)
pd.crosstab(df.stroke,df.work_type,margins=True)
```

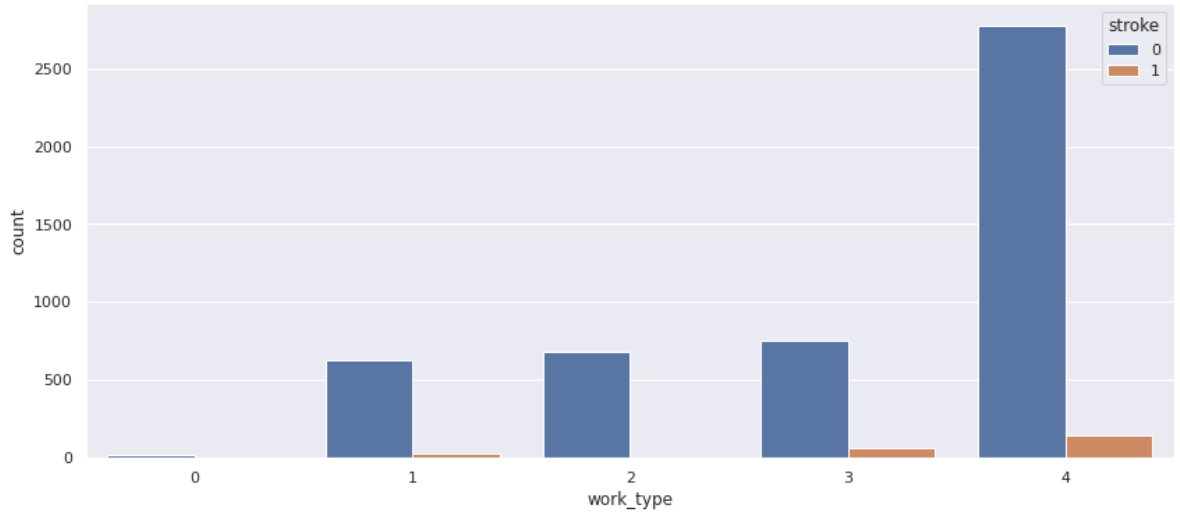
work_type	0	1	2	3	4	All
stroke						
0	22	624	685	754	2776	4861
1	0	33	2	65	149	249
All	22	657	687	819	2925	5110

```
#Layer 2 Pecentage Cross Tab Stroke vs work_type  (C)
pd.crosstab(df.stroke,df.work_type, normalize='index')
```

work_type	0	1	2	3	4
stroke					
0	0.004526	0.128369	0.140918	0.155112	0.571076
1	0.000000	0.132530	0.008032	0.261044	0.598394

```
#Layer 2 (C)
print(df['work_type'].value_counts())
sns.countplot(df['work_type'], hue = df['stroke'])

#We don't have much data on patients that never worked but the data we have in that category never suffered from a stroke. Also the
#  Its obvious that rarely children have stroke. The work type variable by itself does not tell much about the stroke
```

```
4    2925
3     819
2     687
1     657
0      22
Name: work_type, dtype: int64

<AxesSubplot:xlabel='work_type', ylabel='count'>
```



BIVARIATE ANALYSIS

Stroke vs Residence_type

```
#Layer 2 Cross Tab Stroke vs Residence_type (C)
pd.crosstab(df.stroke,df.Residence_type ,margins=True)
```

Residence_type	0	1	All
stroke			
0	2400	2461	4861
1	114	135	249
All	2514	2596	5110

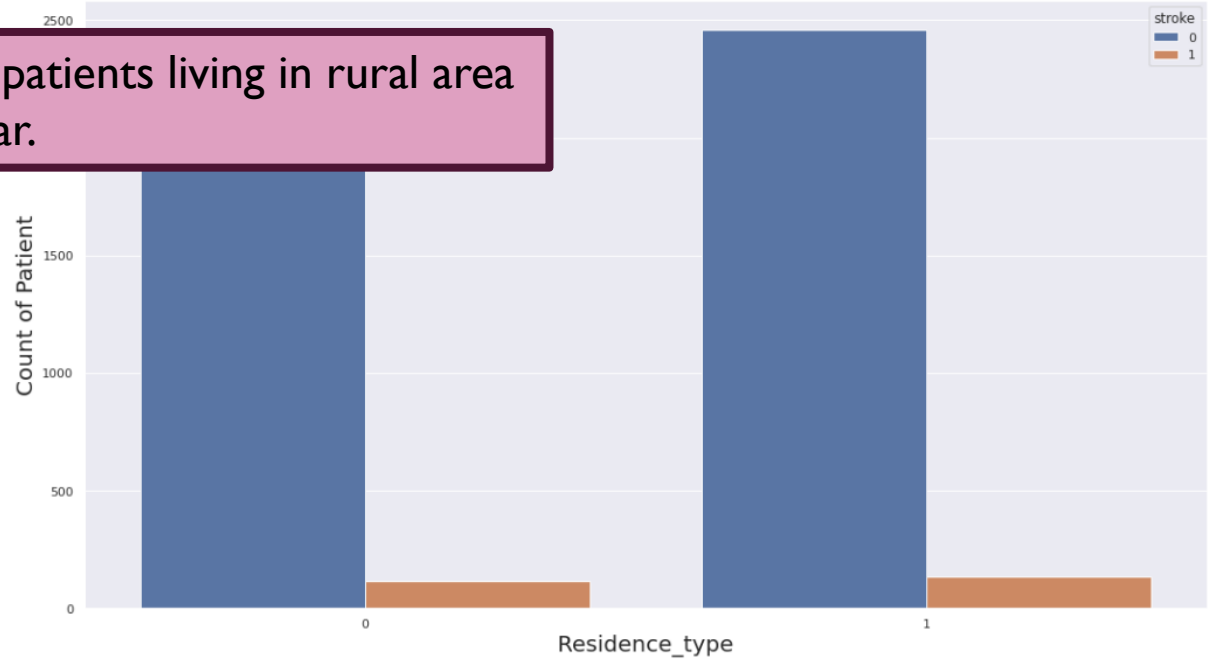
```
#Layer 2 Percentage Cross Tab Stroke vs Residence_type (C)

pd.crosstab(df.stroke,df.Residence_type, normalize='index')
```

Residence_type	0	1
stroke		
0	0.493726	0.506274
1	0.457831	0.542169

```
#Layer 2 (C)
print(df['Residence_type'].value_counts())
sns.set(rc={'figure.figsize':(18,10)})
seaborn_plot = sns.countplot(df['Residence_type'], hue = df['stroke'])
seaborn_plot.set_xlabel("Residence_type",fontsize=20)
seaborn_plot.set_ylabel("Count of Patient",fontsize=20)
```

```
1    2596
0    2514
Name: Residence_type, dtype: int64

Text(0, 0.5, 'Count of Patient')
```

The number of stroke patients living in rural area and urban area is similar.



BIVARIATE ANALYSIS

Stroke vs Smoking_status

```
#Layer 2 Cross Tab Stroke vs smoking_status (C)
pd.crosstab(df.stroke,df.smoking_status,margins=True)
```

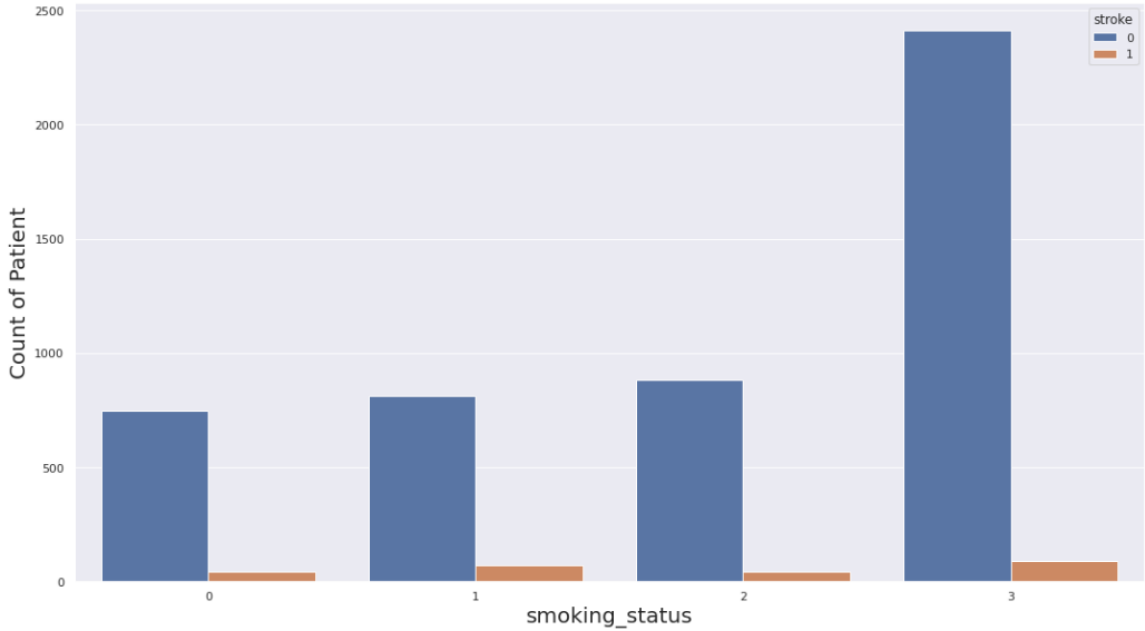
smoking_status	0	1	2	3	All
stroke					
0	747	815	885	2414	4861
1	42	70	45	92	249
All	789	885	930	2506	5110

```
##Layer 2 Percentage Cross Tab Stroke vs smoking status (C)
pd.crosstab(df.stroke,df.smoking_status, normalize='index')
```

smoking_status	0	1	2	3
stroke				
0	0.153672	0.167661	0.182061	0.496606
1	0.168675	0.281124	0.180723	0.369478

```
#Layer 2 Plot smoking_status (C)
print(df['smoking_status'].value_counts())
sns.set(rc={'figure.figsize':(18,10)})
seaborn_plot = sns.countplot(df['smoking_status'], hue = df['stroke'])
seaborn_plot.set_xlabel("smoking_status",fontsize=20)
seaborn_plot.set_ylabel("Count of Patient",fontsize=20)
```

```
3    2506
2     930
1     885
0     789
Name: smoking_status, dtype: int64

Text(0, 0.5, 'Count of Patient')
```



BIVARIATE ANALYSIS

Stroke vs Age_bins

```
#Layer 2 Cross tab Stroke-Age bins (Y)
pd.crosstab(df.stroke, df.age_bins,margins=True)
```

age_bins	0-4.99	5-14.99	15-29.9	30-44.99	45-59.99	60-74.99	75-82.99	All
stroke								
0	254	443	816	1010	1143	779	416	4861
1	1	1	0	8	58	79	102	249
All	255	444	816	1018	1201	858	518	5110

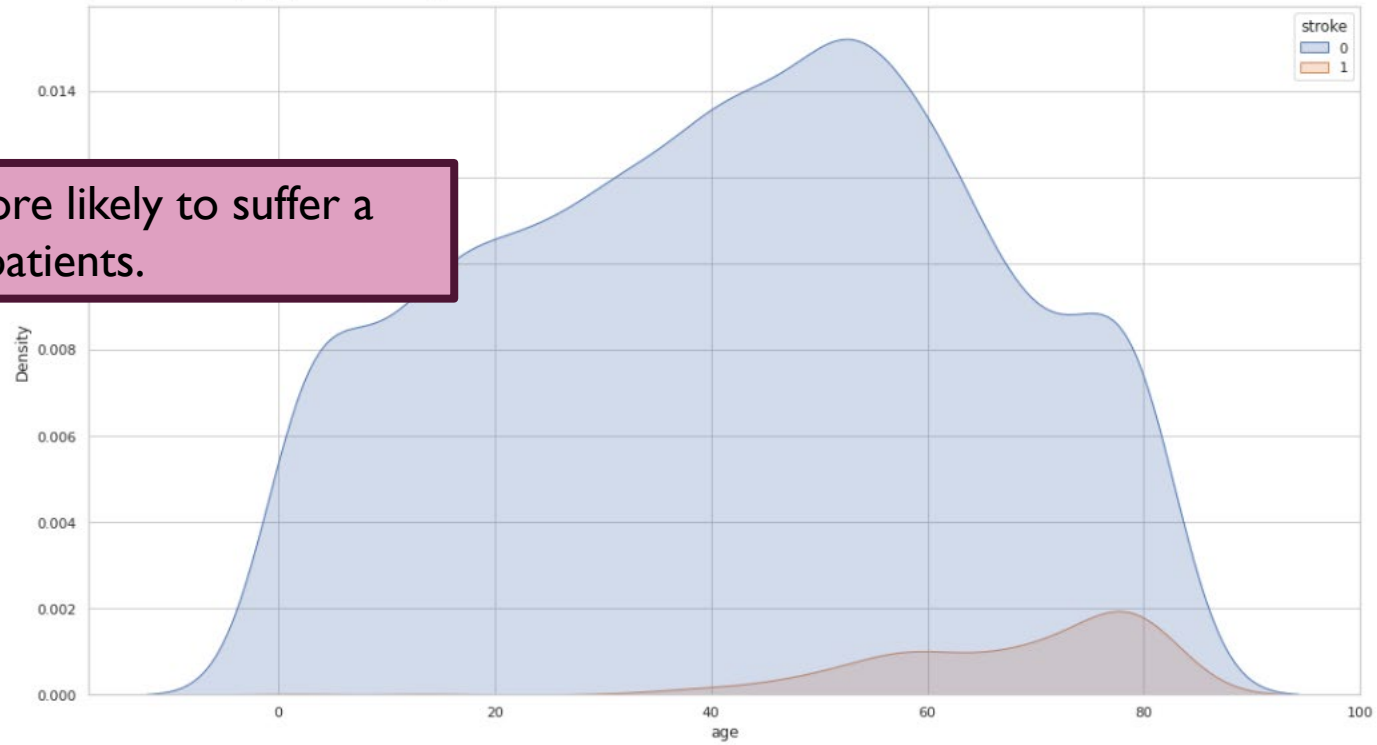
```
#Layer 2 Percentage of Cross Tabulation Stroke vs Age_bins(Y)
all=pd.crosstab(df.stroke, df.age_bins,margins=True)["All"] #deviding by all
pd.crosstab(df.stroke, df.age_bins).divide(all,axis=0).dropna()
```

age_bins	0-4.99	5-14.99	15-29.9	30-44.99	45-59.99	60-74.99	75-82.99
stroke							
0	0.052253	0.091134	0.167867	0.207776	0.235137	0.160255	0.085579
1	0.004016	0.004016	0.000000	0.032129	0.232932	0.317269	0.409639

Older patients are more likely to suffer a stroke than younger patients.

```
# Kernel Distribution Estimation Plot which depicts the probability density function of the continuous (Y)
#or non-parametric data variables
sns.kdeplot(data=df, x="age", hue="stroke", fill=True)
```

```
<AxesSubplot:xlabel='age', ylabel='Density'>
```



BIVARIATE ANALYSIS

Stroke vs Avg_glucose_level

```
#Layer 2 Cross tab Stroke vs glucose_labels (Y)
pd.crosstab(df.stroke, df.glucose_labels,margins=True)
```

glucose_labels	1	2	3	All
stroke				
0	3019	961	881	4861
1	112	37	100	249
All	3131	998	981	5110

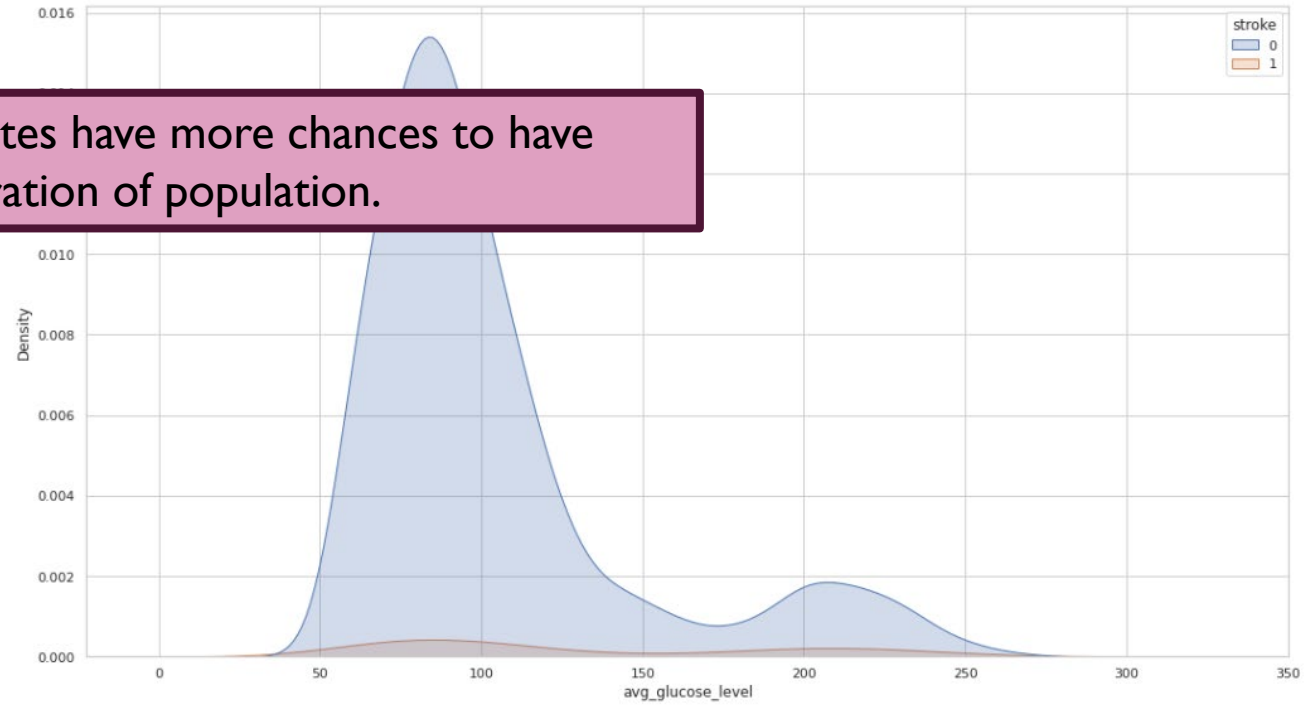
```
#Layer 2 Percentage of Cross Tabulation Stroke vs glucose_labels(Y)
all=pd.crosstab(df.stroke, df.glucose_labels,margins=True)["All"] #deviding by all
pd.crosstab(df.stroke, df.glucose_labels).divide(all,axis=0).dropna()
```

glucose_labels	1	2	3
stroke			
0	0.621066	0.197696	0.181238
1	0.449799	0.148594	0.401606

The patients with diabetes have more chances to have stroke in terms of the ration of population.

```
# Kernel Distribution Estimation Plot (Y)
sns.kdeplot(data=df, x="avg_glucose_level", hue="stroke", fill=True)
```

```
<AxesSubplot:xlabel='avg_glucose_level', ylabel='Density'>
```



BIVARIATE ANALYSIS

Stroke vs BMI

```
#Layer 2 Cross tab Stroke vs bmi_labels (Y)
pd.crosstab(df.stroke, df.bmi_labels,margins=True)
```

bmi_labels	1	2	3	4	All
stroke					
0	341	1224	1334	1962	4861
1	2	35	75	137	249
All	343	1259	1409	2099	5110

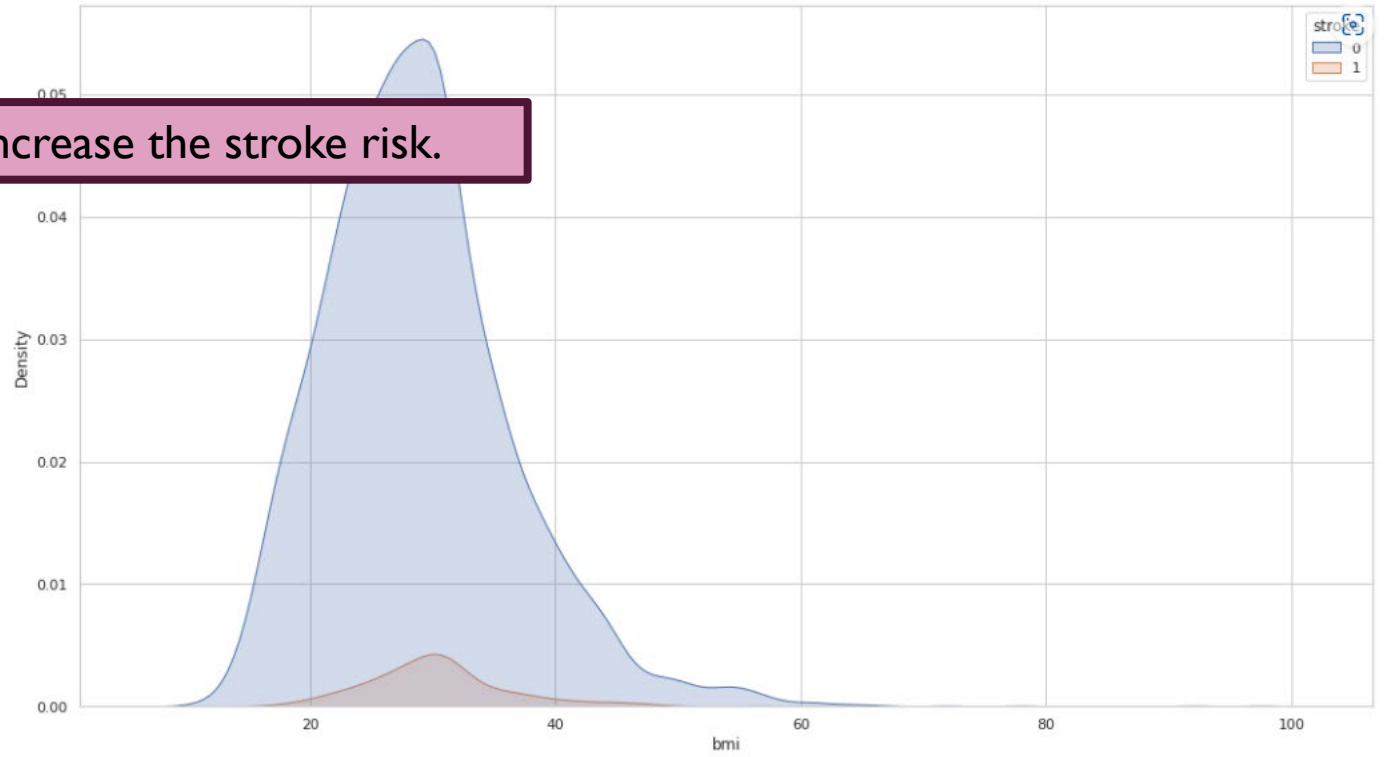
```
#Layer 2 Percentage of Cross Tabulation Stroke vs bmi_labels(Y)
all=pd.crosstab(df.stroke, df.bmi_labels,margins=True)["All"] #deviding by a
pd.crosstab(df.stroke, df.bmi_labels).divide(all,axis=0).dropna()
```

bmi_labels	1	2	3	4
stroke				
0	0.070150	0.251800	0.274429	0.403621
1	0.008032	0.140562	0.301205	0.550201

Higher BMI does not increase the stroke risk.

```
# Kernel Distribution Estimation Plotb (Y)
sns.kdeplot(data=df, x="bmi", hue="stroke", fill=True)
```

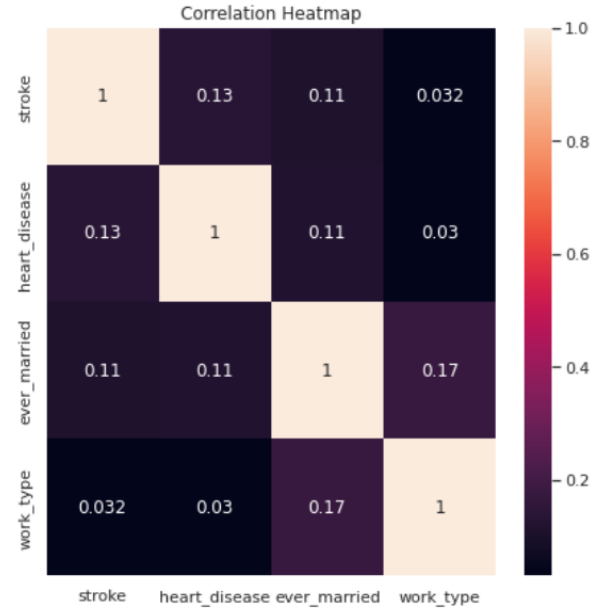
```
<AxesSubplot:xlabel='bmi', ylabel='Density'>
```



CORRELATION

```
#Basic Seaborn Heatmap for 2 continious vsariabels
# The Pearson correlation coefficient can range from -1 to 1
fig, ax = plt.subplots(figsize=(7, 7))
heatmap = sns.heatmap(df[["stroke","heart_disease","ever_married","work_type"]].corr(), vmax=1, annot=True,ax = ax)
heatmap.set_title('Correlation Heatmap')
'''Correlation  0.9 and 1.0 very highly correlated.
Correlation 0.7 and 0.9  highly correlated.
Correlation 0.5 and 0.7 moderately correlated.
Correlation 0.3 and 0.5 low correlation.
Correlation less than 0.3 have little if any (linear) correlation.'''
```

```
Text(0.5, 1.0, 'Correlation Heatmap')
```



T-TEST

```
0.1s
#T-test Gender and Glucose level. The null hypothesis (H0) is that the true difference between these group means is zero.
x, y = df[df.gender=='Male'].age, df[df.gender=='Female'].age
ALPHA=0.05
t, p = stats.ttest_ind(x, y, equal_var=False)
t, p = round(t, 5), round(p, 5)

if p <= ALPHA:
    print(f'Since p = {p} < {ALPHA}, reject H0, different Age distributions between Male and Female (t={t}, p={p})')
else:
    print(f'Since p = {p} > {ALPHA}, fail to reject H0, same Age distributions between Male and Female (t={t}, p={p}).')
```

```
Since p = nan > 0.05, fail to reject H0, same Age distributions between Male and Female (t=nan, p=nan).
```

```
#T-test Gender and Glucose level. The null hypothesis (H0) is that the true difference between these group means is zero.
x, y = df[df.gender=='Male'].avg_glucose_level, df[df.gender=='Female'].avg_glucose_level
ALPHA=0.05
t, p = stats.ttest_ind(x, y, equal_var=False)
t, p = round(t, 5), round(p, 5)

if p <= ALPHA:
    print(f'Since p = {p} < {ALPHA}, reject H0, different avg_glucose_level distributions between Male and Female (t={t}, p={p})')
else:
    print(f'Since p = {p} > {ALPHA}, fail to reject H0, same avg_glucose_level distributions between Male and Female (t={t}, p={p}).
```

```
Since p = nan > 0.05, fail to reject H0, same avg_glucose_level distributions between Male and Female (t=nan, p=nan).
```

T-TEST

```
#T-test Gender and bmi. The null hypothesis (H0) is that the true difference between these group means is zero.
x, y = df[df.gender=='Male'].bmi, df[df.gender=='Female'].bmi
ALPHA=0.05
t, p = stats.ttest_ind(x, y, equal_var=False)
t, p = round(t, 5), round(p, 5)

if p <= ALPHA:
    print(f'Since p = {p} < {ALPHA}, reject H0, different BMI distributions between Male and Female (t={t}, p={p})')
else:
    print(f'Since p = {p} > {ALPHA}, fail to reject H0, same BMI distributions between Male and Female (t={t}, p={p}).')
```

```
Since p = nan > 0.05, fail to reject H0, same BMI distributions between Male and Female (t=nan, p=nan).
```

```
#bivariate analysis
#   cat_feat    num_feat    alpha            t              p         H0           relation
#0  gender  age                0.050000     -1.961000      0.049900    Rejected     Different
#1  gender  avg_glucose_level   0.050000    3.860000       0.000120    Rejected     Different
#2  gender  bmi                0.050000    -1.921000       0.054780    Fail to rej  same

# It is summarized clearly in table forms here, where only the distributions of gender and bmi are the same.
```

CHI-SQUARE TEST

Stroke Vs Work_type

```
#CHI-SQUARE STROKE and WORK TYPE
myF1= df['stroke']
myF2= df['work_type']
mycrosstable= pd.crosstab(myF2,myF1)
print(mycrosstable)
chiVal, pVal, df1, exp= chi2_contingency(mycrosstable)
chiVal, pVal, df1, exp
```

```
stroke        0    1
work_type
0            22    0
1           624   33
2           685    2
3           754   65
4          2776  149

(49.16351197667529,
 5.397707801896138e-10,
 4,
 array([[2.09279843e+01, 1.07201566e+00],
        [6.24985714e+02, 3.20142857e+01],
        [6.53523875e+02, 3.34761252e+01],
        [7.79091781e+02, 3.99082192e+01],
        [2.78247065e+03, 1.42529354e+02]]))
```

Chi-square(49.16), p-value (5.39e-10), degrees of freedom (4), expected frequencies are less than 5=> we can not full trust this test, reject H0=>there is a relationship.

Stroke Vs Age_bins

```
#CHI-SQUARE STROKE and AGE_BINS
myF1= df['stroke']
myF2= df['age_bins']
mycrosstable= pd.crosstab(myF2,myF1)
print(mycrosstable)
chiVal, pVal, df1, exp= chi2_contingency(mycrosstable)
chiVal, pVal, df1, exp
```

```
stroke       0    1
age_bins
0-4.99     254    1
5-14.99    443    1
15-29.9    816    0
30-44.99  1010    8
45-59.99  1143   58
60-74.99   779   79
75-82.99   416  102

(390.3822065889208,
 3.265283011373326e-81,
 6,
 array([[ 242.57436399,   12.42563601],
        [ 422.36477495,   21.63522505],
        [ 776.23796477,   39.76203523],
        [ 968.39491194,   49.60508806],
        [1142.4776908 ,   58.5223092 ],
        [ 816.19138943,   41.80861057],
        [ 492.75890411,   25.24109589]]))
```

Chi-square(390.382), p-value (3.26e-81), degrees of freedom (6), expected frequencies are greater than 5=>we can trust results, reject H0=>there is a relationship.

CHI-SQUARE TEST

Stroke Vs Hypertension

```
myF1= df['stroke']
myF2= df['hypertension']
mycrosstable= pd.crosstab(myF2,m
print(mycrosstable)
chiVal, pVal, df1, exp= chi2_con
chiVal, pVal, df1, exp
```

```
stroke            0    1
hypertension
0              4429  183
1               432   66

(81.6053682482931,
 1.661621901511823e-19,
 1,
 array([[4387.2665362,  224.7334638],
        [ 473.7334638,   24.2665362]]))
```

Chi-square(81.6), p-value (1.66e-19), degrees of freedom (1), expected frequencies are greater than 5=> we can trust results, reject H0=>there is a relationship.

Stroke Vs Heart_disease

```
myF1= df['stroke']
myF2= df['heart_disease']
mycrosstable= pd.crosstab(myF2,myF1)
print(mycrosstable)
chiVal, pVal, df1, exp= chi2_c
chiVal, pVal, df1, exp
```

```
stroke             0    1
heart_disease
0               4632  202
1                229   47

(90.25956125843324,
 2.0887845685229236e-21,
 1,
 array([[4598.44892368,  235.55107632],
        [ 262.55107632,   13.44892368]]))
```

Chi-square(90.6), p-value (0.088e-21), degrees of freedom (1), expected frequencies are greater than 5=>we can trust results, reject H0=>there is a relationship.

CHI-SQUARE TEST

Stroke Vs Age Ever_married

```
myF1= df['stroke']
myF2= df['ever_married']
mycrosstable= pd.crosstab(myF2,myF1)
print(mycrosstable)
chiVal, pVal, df1, exp= chi2_contingency(mycrosstable)
chiVal, pVal, df1, exp
```

```
stroke            0    1
ever_married
0              1728   29
1              3133  220

(58.923890259034195,
 1.6389021142314745e-14,
 1,
 array([[1671.38493151,   85.61506849],
        [3189.61506849,  163.38493151]]))
```

Chi-square(58.92), p-value (1.6e-14), degrees of freedom (1), expected frequencies are greater than 5=> we can trust results, reject H0=>there is a relationship.

Stroke Vs Residence_type

```
myF1= df['stroke']
myF2= df['Residence_type']
mycrosstable= pd.crosstab(myF2,myF1)
print(mycrosstable)
chiVal, pVal, df1, exp= chi2_
chiVal, pVal, df1, exp
```

```
stroke              0    1
Residence_type
0                2400  114
1                2461  135

(1.0816367471627524,
 0.29833169286876987,
 1,
 array([[2391.49784736,  122.50215264],
        [2469.50215264,  126.49784736]]))
```

Chi-square(1.08), p-value (0.29), degrees of freedom (1), expected frequencies are greater than 5=>we can trust results, fail to reject H0=>there is a relationship.

CHI-SQUARE TEST

Stroke Vs Glucose_labels

```
myF1= df['stroke']
myF2= df['glucose_labels']
mycrosstable= pd.crosstab(myF2,myF1)
print(mycrosstable)
chiVal, pVal, df1, exp= chi2_c
chiVal, pVal, df1, exp
```

```
stroke              0    1
glucose_labels
1                3019  112
2                 961   37
3                 881  100

(74.18076312175809,
 7.795643250529761e-17,
 2,
 array([[2978.43268102,  152.56731898],
        [ 949.36947162,   48.63052838],
        [ 933.19784736,   47.80215264]]))
```

Chi-square(74.18), p-value (7.779e-17), degrees of freedom (2), expected frequencies are greater than 5=> we can trust results, fail to reject H0=>there is a relationship.

Stroke Vs Bmi_labels

```
myF1= df['stroke']
myF2= df['bmi_labels']
mycrosstable= pd.crosstab(myF2,myF1)
print(mycrosstable)
chiVal, pVal, df1, exp= chi2_contingen
chiVal, pVal, df1, exp
```

```
stroke          0    1
bmi_labels
1             341    2
2            1224   35
3            1334   75
4            1962  137

(38.51824910812336,
 2.1953680478046033e-08,
 3,
 array([[ 326.28630137,   16.71369863],
        [1197.65146771,   61.34853229],
        [1340.34227006,   68.65772994],
        [1996.71996086,  102.28003914]]))
```

Chi-square(38.51), p-value 2.19e-08), degrees of freedom (1), expected frequencies are greater than 5=> we can trust results, fail to reject H0=>there is a relationship.

CHI-SQUARE TEST

Stroke Vs Smoking_status

```
myF1= df['stroke']
myF2= df['smoking_status']
mycrosstable= pd.crosstab(myF2,myF1)
print(mycrosstable)
chiVal, pVal, df1, exp= chi2_contingency(mycrosstable)
chiVal, pVal, df1, exp
```

```
stroke             0   1
smoking_status
0                747  42
1                815  70
2                885  45
3               2414  92

(25.760905375925446,
 1.0702431639507865e-05,
 3,
 array([[ 750.55362035,   38.44637965],
        [ 841.87573386,   43.12426614],
        [ 884.68297456,   45.31702544],
        [2383.88767123,  122.11232877]]))
```

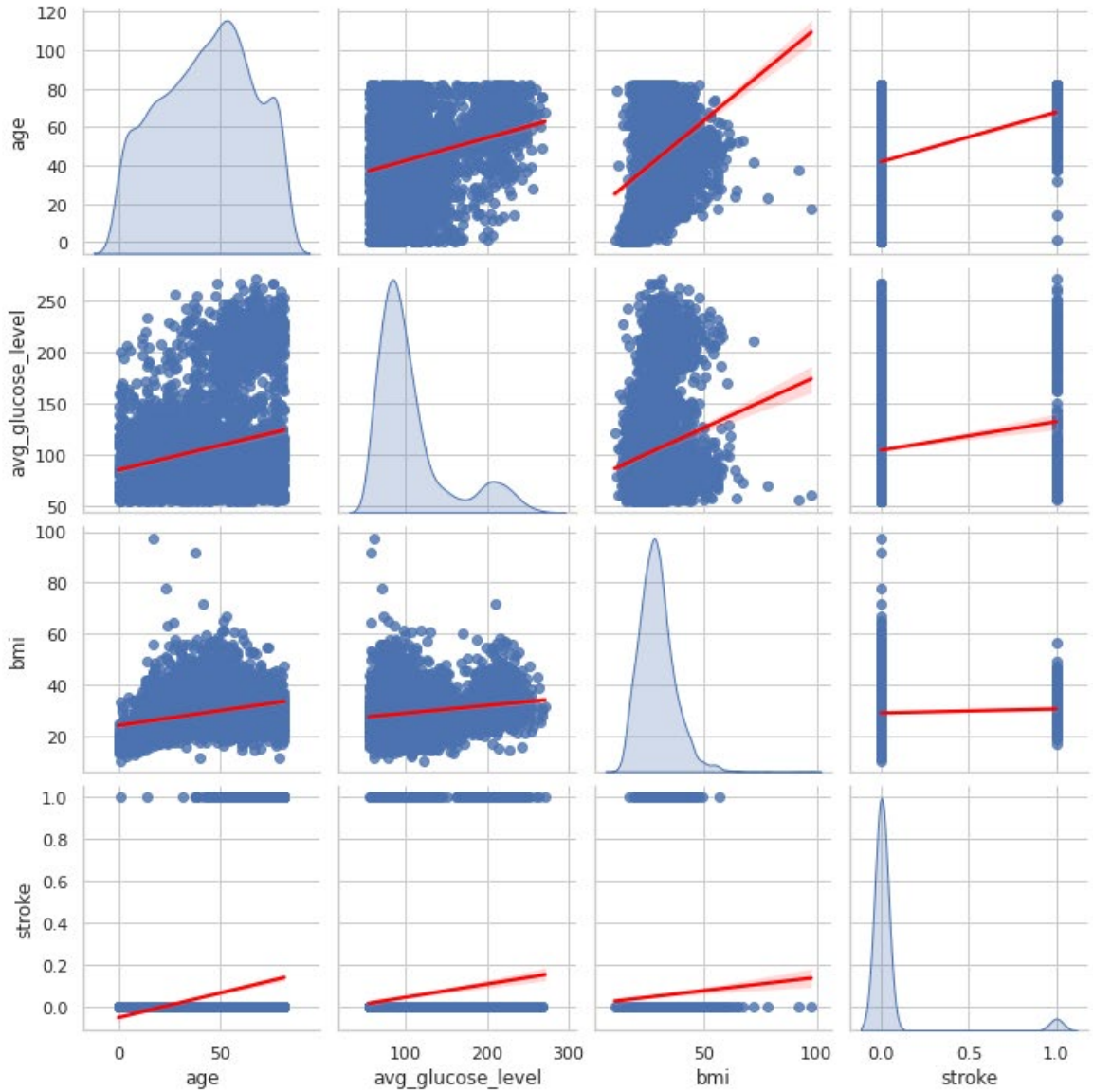
Chi-square(25.76), p-value (1.07e-05), degrees of freedom (3), expected frequencies are greater than 5=>we can trust results, fail to reject H0=>there is a relationship.

Percentage of each glucose_label whoever suffered stroke:1-normal 3.6%2-prediabetic 3.9%3-diabetic 10.2%We see that percentage of patiens with diabetics who suffered stroke has the highest value.We can conclude that the first greatest dependancy is between stroke and age (Chi_square=390.382).The second is stroke heart_disease with Chi-square value of 90.6. And the third is stroke and hypertension (Chi=square=81.6).The forth greatest dependancy is stroke and glucose level with the highest percentage (10.2%) of patiens who suffered stroke falls into diabetic group.The least dependancy for the stroke and smokins_status.

REGRESSION

```
sns.pairplot(df[["age","avg_glucose_level","bmi","stroke"]], diag_kind='kde', kind='reg',plot_kws={'line_kws':{'color':'red'}})
```

```
<seaborn.axisgrid.PairGrid at 0x7f87328ca910>
```

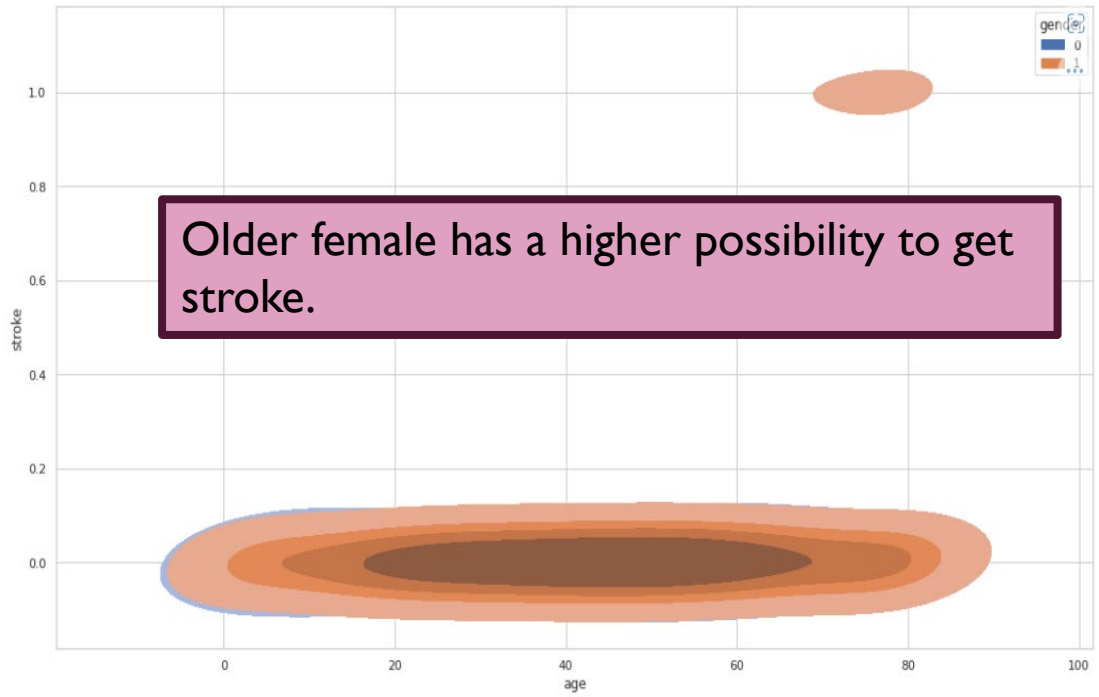


MULTIVARIATE

Stroke vs Gender vs Age

```
sns.kdeplot(data=df, x="age", y='stroke', hue="gender", fill=True,levels=5,thresh=0.1)
```

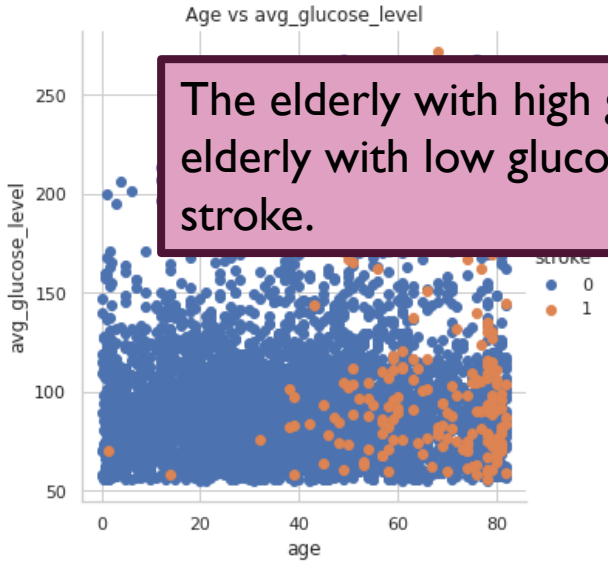
```
<AxesSubplot:xlabel='age', ylabel='stroke'>
```



Older female has a higher possibility to get stroke.

Stroke vs Gluecose vs Age

```
sns.set_style("whitegrid")
sns.FacetGrid(df, hue="stroke", height=5).map(plt.scatter, "age", "avg_glucose_level").add_legend()
plt.title('Age vs avg_glucose_level')
plt.show()
```



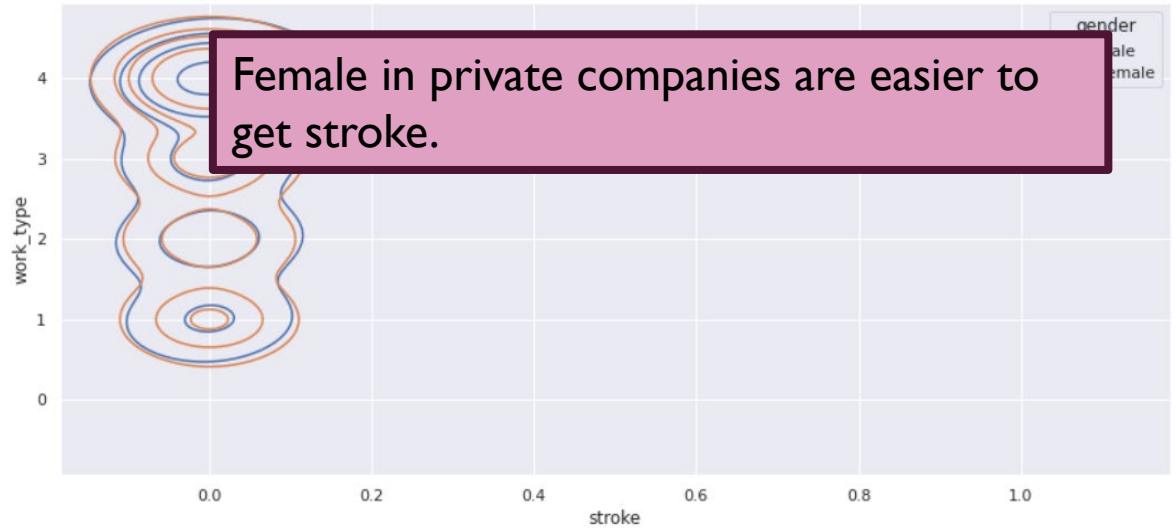
The elderly with high glucose and the elderly with low glucose are both prone to stroke.

MULTIVARIATE

Stroke vs Gender vs Work_type

```
df['work_type'].replace(['Never_worked', 'Govt_job', 'children', 'Self-employed', 'Private'],
                        [0, 1, 2, 3, 4], inplace=True)

sns.kdeplot(data=df, x="stroke", y='work_type', hue="gender", fill=False,levels=5,thresh=0.1)
```

```
<AxesSubplot:xlabel='stroke', ylabel='work_type'>
```

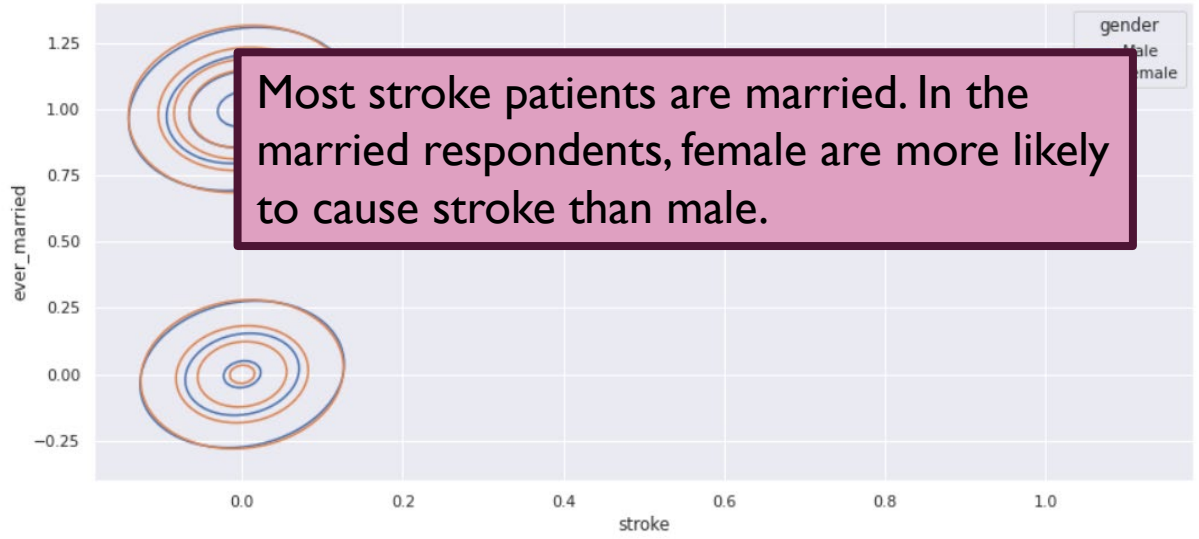


Female in private companies are easier to get stroke.

Stroke vs Gender vs Ever_married

```
df['ever_married'].replace(['No', 'Yes'],
                           [0, 1], inplace=True)
sns.kdeplot(data=df, x="stroke", y='ever_married', hue="gender", fill=False,levels=5,thresh=0.1)
```

```
<AxesSubplot:xlabel='stroke', ylabel='ever_married'>
```

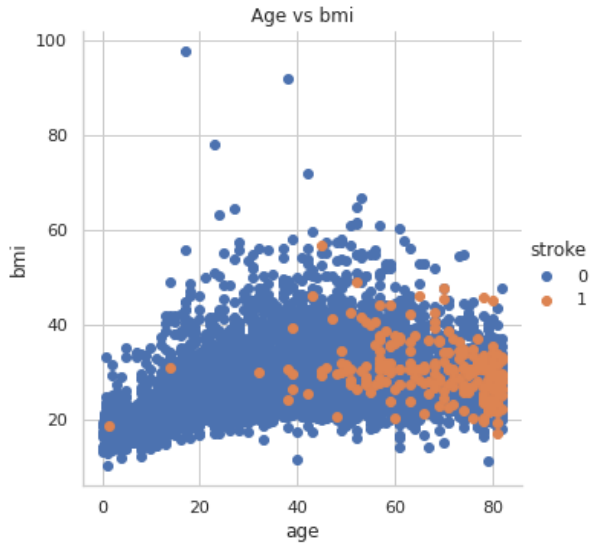


Most stroke patients are married. In the married respondents, female are more likely to cause stroke than male.

MULTIVARIATE

Stroke vs BMI vs Age

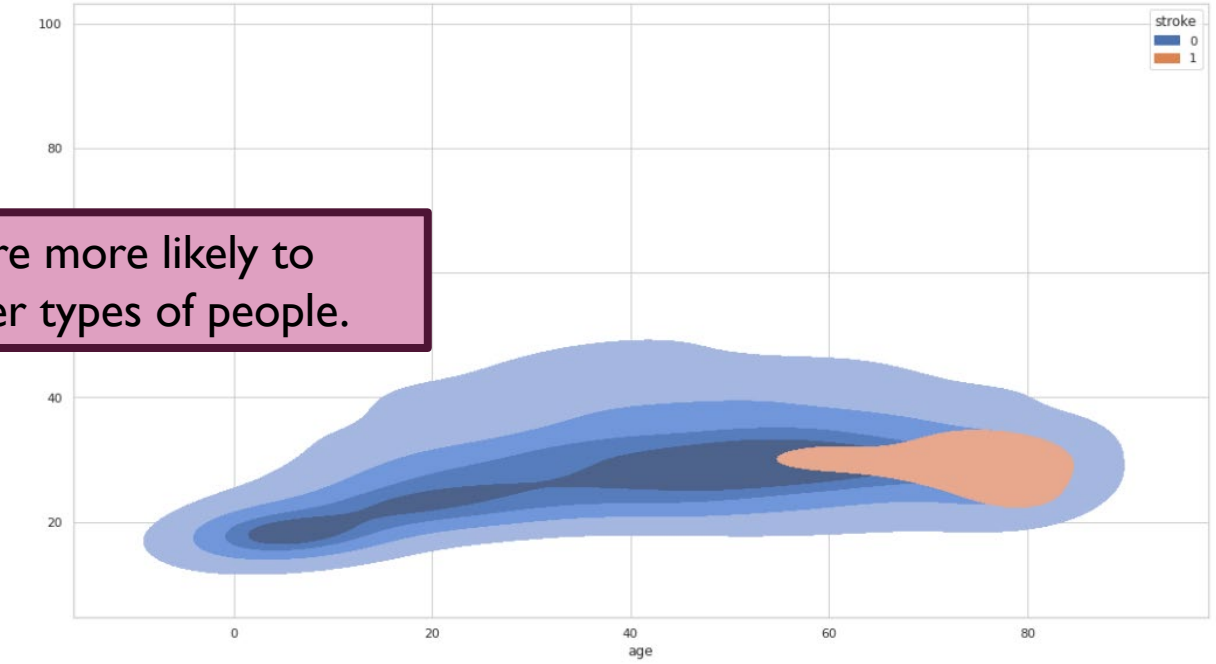
```
#Avg_glucose vs stroke vs gender #Layer 2 (Y)
#  Layer 2 Bivariate analysis / avg_glu_level,age,stroke ASMA
sns.set_style("whitegrid")
sns.FacetGrid(df, hue="stroke", height=5).map(plt.scatter, "age", "bmi").add_legend()
plt.title('Age vs bmi')
plt.show()
```



Elderly obese people are more likely to have a stroke than other types of people.

```
sns.kdeplot(data=df, x="age", y='bmi', hue="stroke", fill=True,levels=5,thresh=0.1)
```

```
<AxesSubplot:xlabel='age', ylabel='bmi'>
```

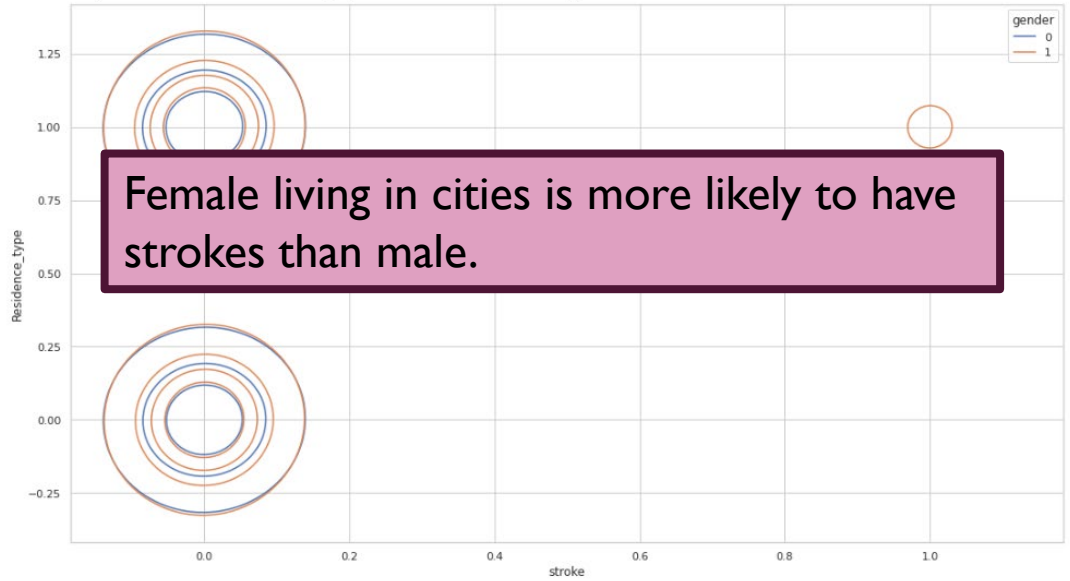


MULTIVARIATE

Stroke vs Gender vs Residence_type

```
.kdeplot(data=df, x="stroke", y='Residence_type', hue="gender", fill=False,levels=5,thresh=0.
```

```
<AxesSubplot:xlabel='stroke', ylabel='Residence_type'>
```

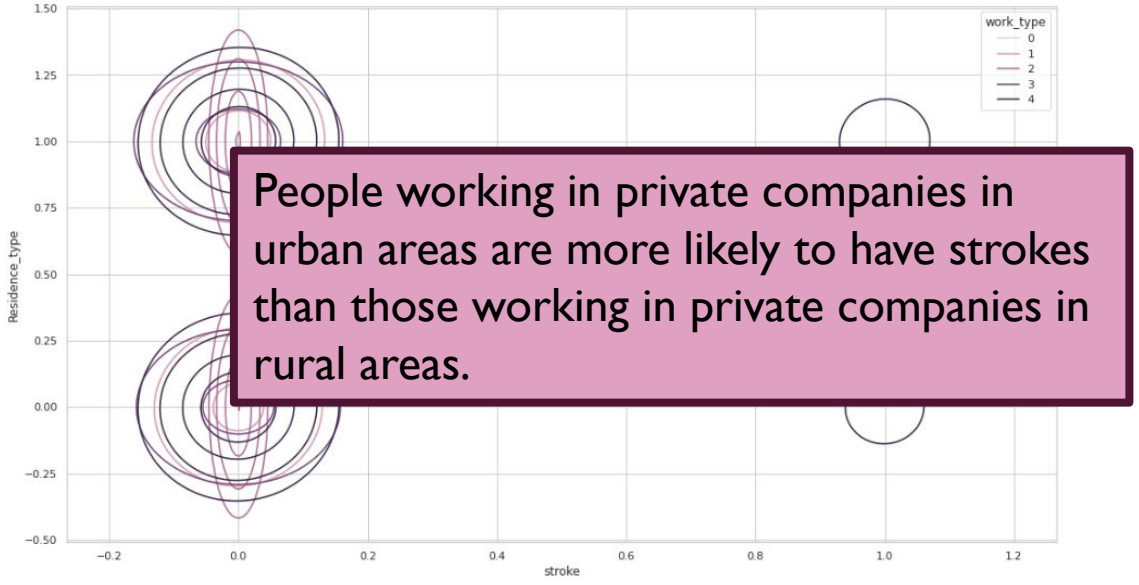


Female living in cities is more likely to have strokes than male.

Stroke vs Residence_type vs Work_type

```
deplot(data=df, x="stroke", y='Residence_type', hue="work_type", fill=False,levels=5,thresh=0
```

```
<AxesSubplot:xlabel='stroke', ylabel='Residence_type'>
```



People working in private companies in urban areas are more likely to have strokes than those working in private companies in rural areas.

LOGISTIC REGRESSION

Logistic Regression to Predict Stroke Using Different Variables

```
#Logistic Regression to predict stroke using different variables
training_df=df[['age', 'avg_glucose_level','bmi' , 'stroke']]
y= training_df['stroke']
x= training_df.drop(["stroke"],axis=1,inplace=False)
x_train,x_test,y_train,y_test=train_test_split(x,y,test_size=0.25,random_state=16)


lr = LogisticRegression()

# fit the model with data
lr.fit(x_train,y_train)
test_predict=lr.predict(x_test)

print("predicted strokes" ,test_predict)
print(lr.predict([[45,100,30],[80,100,200],[20,56,18],[45,93.72,30.2],[66,151.16,27.5],[68,211.06,39.3],[100,402,80]]))

score = lr.score(x_test, y_test)
print("accuracy " ,score)
```

```
predicted strokes [0 0 0 ... 0 0 0]
[0 0 0 0 0 0 1]
accuracy  0.9491392801251957
```

CONCLUSION

- 1. It seemed like both BMI and Age were positively correlated, though the association was not strong.
- 2. Older patients are more likely to suffer a stroke than a younger patient. And elderly obese people are more likely to have a stroke than other types of people.
- 3. Higher BMI does not increase the stroke risk.
- 4. The elderly with high glucose and the elderly with low glucose are both prone to stroke.
- 5. Female in private companies are easier to get stroke.
- 6. Most stroke patients are married. In the married population, women are more likely to cause stroke than men.
- 7. Female living in cities is more likely to have strokes than male.
- 8. People working in private companies in urban areas are more likely to have strokes than those working in private companies in rural areas.

Thank you !